COSTED SCHEDULE OF WORK

Name of Work : Provn of qty 01 x Living Shelter (ORL) D/S at 56 Inf Div AoR

Job No : OT/EC/3/56/2026-27/2653

Ser No	Description of work		Brief Specifications	A/U	Qty
1	Provn of qty 01 x Living Shelter (ORL) (D/S)**(Shelter Part , Constr Mtrls including E/M Items)**		Provn of qty 01 x of Pre-Engineered Portable Frame OR Living (Double Storey) PIR Shelter of size 18.0m x 6.1m with 1.5m wide verandah including roof covered steel staircase with SS handrails on one side of the shelter as in drawing. The structure made out of primary members of High Tensils Steel (340 Mpa min) and Secondary members of (240 Mpa) steel. The main wall and portion and partition wall if any, will be made up of **60mm thick** PIR Panels. The Mezzainine Floor shall be with 1.0 mm Thick Galvanized Deck sheet. Layout of partition wall of ground and first floors. if any, will be as per the respective floor plans attached. The main beam of stair case are made up of built up C section of size 75mm, 200mm, 8/10mm and step platform are mae up of chequered plate 4mm thick. The side railings are made up of square hollow steel sections of 32mm x 32mm x 2.9mm @ spacing not greater than 300mm. The roofing will be made of 0.5mm PPGI colour coated sheets. Detailed specification of shelter and constr part as given in **Appx 'A'** and **Appx 'B'** att. Store List of elect items and constr mtrl are att as per **Annx - I & II.**	Set	1.00
2	Provn of qty 01 x Living Shelter (ORL) (D/s) **(Pathway)(10.00 x 1.20 mtr)**		Provn of interlocking paving blocks (size 10.00x 1.200 mtr) for pathway as per detailed specifications given as per **Appx 'C'** and store list as per **Annx - III.**	Job	1.00
3	Provn of qty 01x Living Shelter (ORL) **(Furniture Part)**	5.1	Provn of **Single Sleeping Bunk with Coir Mattress** as per tech specification given in **Appx 'D'**	Nos	40.00
		5.2	Provn of **Bed Side Locker (Small)** as per tech specification given in **Appx 'E'**	Nos	40.00
		5.3	Provn of **Chair Writing Wooden** as per tech specification given in **Appx 'F'**	Nos	4.00
		5.4	Provn of **Writing Table** as per tech specification given in **Appx 'F'**	Nos	4.00
		5.5	Provn of **water Dispencer** all as per tech specifications given in Appx 'F'	Nos	2.00
		5.6	Provn of Powder based **fire extinguisher** (Red) - 1 as per tech specification given in **Appx 'F'**.	Nos	4.00

Appx 'A'
(Ref Ser No 1 of CS to TS No -1 of
Job No OT/EC/3/56/28-27/2653

DETAILED TECHNICAL SPECIFICATION OF OR LIVING SHELTER D/S OF SIZE (18 M X 6.1M WITH 1.5 M WIDE VERANDAH) : SHELTER PART

1. Design, supply & erection of Pre-Engineered portal frame **Double Storey PIR Shelter** of size 18.0 m x 6.1 m with 1.5 m wide verandah including roof covered steel staircase with SS handrails on one side of the shelter as in drawing. The structure made of primary members of High Tensile Steel (340 Mpa min) Yield Strength and Secondary members of (240 Mpa) Yield Strength of steel. The main wall and partition wall if any, will be made up of **60 mm** thick PIR Panels. The Mezzanine Floor shall be with 1.0 mm Thick Galvanized Deck sheet as per specification attached. Layout of partition walls of ground and first floors, if any, will be as per the respective floor plans attached. The main beam of stair case are made up of built up C section of size 75 mm, 200 mm, 8/10 mm and step platform are made up of chequered plate 4 mm thick. The side railings are made up of square hollow steel sections of 32 mm x 32 mm x 2.9 mm @ spacing not greater than 300 mm. The roofing will be made of 0.5mm PPGI colour coated sheets. Other specifications are as per description given under each specification and drawings Nos DSB - 01 to 03 attached.

2. **Design Parameters.**

(2.1)	Seismic Co-efficient	-	As per Seismic Zone VI.
(2.2)	Wind Load	-	Equivalent to wind speed of 55m/sec as per IS-875.
(2.3)	Roof Slope	-	1:3
(2.4)	External Temperature	-	(-)10°C to (+)50°C
(2.5)	Water Absorption/ Penetration	-	Nil
(2.6)	Termite Proofing	-	Termite Proof.
(2.7)	Fire Resistant	-	Fire retardant and should not emit toxic fumes.
(2.8)	Ease of Constr	-	Shelter should be modular in design.
(2.9)	Design Load	-	As per IS- 875.

3. **Dimensions.**

(3.1)	Length	-	18.00 Mtr.
(3.2)	Width	-	6.10 Mtr.
(3.3)	Verandah width	-	1.50 Mtr.
(3.4)	Internal height of each Storey upto false ceiling	-	3.66 Mtr.

4. **Details of Structural Members.** The steel structure for the shelters shall be designed for the design parameters as enumerated in Para 2 and for the dimensions indicated in Para 3 and drawings attached. The detailed design calculations, structural drawings and foundation design shall be submitted by the firm duly vetted by chartered structural Engineer/IIT/NIT.

(4.1) Roof members and columns shall be designed to stand max wind speed of 55 m/sec. The steel should be conforming to IS-1161of 1979 or IS 2062-1984.

(4.2) The roof shall be provided with 0.50 mm colour coated trapezoidal PPGI sheet.

(4.3) The roof shall have minimum projection of 0 45 Mtr from the eaves wall. Purlins shall also be provided in the roof at the eaves and at gable roof extension (outside the structure)

(4.4) **Steel Stair Case.** The main beam of stair case made up of built up C section of size 75 mm, 200 mm, 8/10 mm and step platform are made up of chequered plate 4 mm thick. The side railing are made up of square hollow stainless steel sections of 32 mm x 32 mm x 2.9 mm @

spacing not > 300 m. The staircases would be provided overhead roof in continuation with the shelter roof and roof projection of 0.45 m on gable side is also provided. The stair case is opened on all other three sides.

(4.5) **Foundation bolts**. Foundation bolts of size dia 20 mm x 685 mm long. Four numbers for each foundation. The bolts must be made from hard steel.

5. **Roofing**.

(5.1) **Roof**. The roof slope shall be laid in 1:4. Roofing shall be done with roof sheet of PPGI 0.50 mm with guard film. The PPGI sheet shall have minimum coating of 4-5 micron epoxy primer and 18-20 micron polyester top coat. The top of 0.50 mm PPGI will have a guard film of 30-40 micron on the finished surface only for protection against scratches. The Pre coated GI sheet will confirm to IS-14246-1995. Base metal of PPGI will be of CRCA as per IS-513 and galvanised as per IS-277 having zinc coating mass of 120 GSM. The roof shall be laid over a frame work of trusses, purlins and column fixed using tapping screw with steel washer.

(5.2) **Ridge Cover**. Ridge cover of size 225 mm x 225 mm x 0.5 mm made from pre coated plain sheet matching with the roof sheet shall be provided.

(5.3) **Roof projections**. The roof shall have minimum projections 0.45 m for the eaves wall in the rear in front where no verandah is provided and also on both gable walls. Purlins will be provided for the roof projections at the eaves and at gable ends (outside the structure made from MS pipe 80 NB (H) conforming toIS-1161-1779).

(5.4) **Overlapping**. Sheets to have minimum overlap of 150 mm.

6.0 **Wall Panels (Internal/External)-** All walls shall be made of PIR insulted panel with insulation core polyisocyanurate foam (PIR) insulation. **PIR insulated panel shall be 60 mm thick** with **density 40 Kg/Cum or more**. The panel insulation material in the panel shall have fire retarding and self -extinguishing properties as per international standard. There should not be any gap between wall panels and wall and roof by provision of suitable joining arrangement. Other details of wall panels are as follows.

6.1 All material required for the manufacture of shelter shall be new and shall comply with relevant Bureau of Indian Standard Specification.

6.2 The bulk density of PIR panel insulation should be **40 Kg /Cum or more insulation core (PIR) polyisocyanurate foam** and the dispensing machinery should be kept with a PLC controlled panel for monitoring and controlling the injection rate to assure specified uniform density requirements. The total thickness of the finished composite panel should be 51 mm. The tolerance in the panel can only be on the plus side

6.3 The outer and inner colour coated (RAL-1001) HI skin made from hot dipped galvanized steel of the panels should be 0.50 mm thick TCT (Total coated thickness). Each panel should be supplied pre-painted with a colour coating of 50 microns of architectural polyester on a minimum 120 GSM base of Zink coating on the finished surface only for protection against scratches during handling and transportation. Base metal of GI skin CRCA should be as per **IS-513-2016 (reaffirmed 2021)** and grade of zinc coating will be 120 GSM as per IS-277-2018 **(reaffirmed 2022)**

6.4 The pre-coated GI sheet skin should have minimum coating of 4-5 micron epoxy primer and 25 micron polyester top coat on the finished surface and 7-8 micron primer Alkyd back coat which is bounded to the (PIR) polyisocyanurate foam. The pre coated GI sheet should conform to **IS-14246-2013 (reaffirmed 2018)** with manufacture test certificates conforming to specification.

6.5 The PIR insulated core of these composite panels shall have the following properties :-

6.5.1 Density (IS 11239: Part 2) - **40 Kg/ m^3 or more**

6.5.2 Panel wt should not be exceed - Max 11.70 Kg/ m^3

6.5.3 U value of PIR panel will not exceed – 0.30 w/m2K

6.5.4 Flammability - BS2D0(min)

6.5.5 Blowing Agent – Non CFC & Non HCFC

6.5.6 Compressive Strength at 10 % deformation - 115 KN/m^2

6.5.7 Tensile Strength (Adhesion Strength) (IS 16203) – 2.5 Kg/Sqcm

6.5.8 K Value (IS 16203) - Equal or less than 0.023W/mk at 10^0 temp & 0.025 w/mk at 25^0 temp

6.5.9 Water absorption (ISO 2986) – 8-5 ng/Pasm(max)

6.5.10 Close Cell Content (IS 11239 Part 5) – Min 85 %

6.5.11 Air and water permeability – No absorption of vapor or water

6.5.12 Horizontal burning characteristics – max 25 mm

6.5.13 Dimensional Stability at 100^0 C and - 40^0 C Overall max 2.0 % (IS 11239:Part 3)

6.5.14 Core Type – Self blended hybrid insulation core

6.5.15 Composition – Foam based closed cell structure

6.5.16 Green Mtrls – 100 % recyclable

6.5.17 Fire certification – Fire safe test report showing integration, insulation as per EN13501-2

6.5.18 Effect of Fire – No loss of insulation core to be ensured

6.5.19 **Smoke generation**

6.5.19.1 Smoke generation rate at 12 min <0.07g/s

6.5.19.2 Max smoke generation rate <0.23 g/s

6.5.19.3 Total smoke generated < 60 g

6.5.20 Joints in Panel – Joint less (Length upto 11.8 Mtr)

6.5.21 **Ash Content**

6.5.21.1 The core material should have minimum ash content of 90 % when tested without adhesive

6.5.21.2 The core material has max heat combustion of 2.0 kj/g (860 BTU/lb) when tested without adhesive

6.5.21.3 The core material should not show visible flaming when tested at an applied heat flux of 50 kW /m2 in air enriched to 40 % oxygen without adhesive or facers.

6.5.22 **Note :- All panels will be manufactured in single piece as per approved layout drawings using the above materials and manufacturing process. Continuous line Polyisocyanurate foam Panels, being used in wall /ceiling /any other location will be tested for its physical /chemical /mechanical properties. Test Certificate for PIR insulation given by CSIR**

-CBRI Roorkee . Certificate will also give out conformity to the following IS – 12436 -1988 (reaff 2022)IS 11239 part 2,4,5 11 & 12 1985- 1988 (reaff 2019) and flammability test certification conforming EN13501-1 or 476 Part 20 to 22 from CSIR-CBRI Roorkee /IAS labs/NABL will be produced by supplier to consignee at the time of supply of each asset.

6.6 2 mm thick MS Sheet 'C' channel of size 55 mm x 40 mm at the bottom (Floor level) and Top shall be provided to side and fit the wall panels. Suitable arrangement will be made to fix the 'C' channel to floor at bottom and top to structural members.

6.7 All panels should be moulded in place using the above in-situ process through continuous line manufacturing process after placing them in a hydraulic process. A spare panel will be supplied along with bulk supply of panels for testing the designed parameters of the incorporated panels in the shelters.

6.8 The purchaser can get quality testing of any panel from the lot supplied to ensure quality control as per given specifications. Cost of quality testing will be borne by supplier.

7. **Locking arrangement.** All PIR panels will be provided with suitable number of cam-lock which are fixed in place during the in situ process itself. The placing of cam-lock should be eccentric and such that a 1.2 m long x 1.2 m wide wall panel should be provided with three pairs of cam-lock of the vertical joints. The holes in the panel for operating the cam-lock should be provided with PVC caps for sealing after installations.

8. **Gable End.** PPGI metal skin PIR panels of 50 mm. shall be provided at both gable ends excluding the stair case of the shelters with necessary fixing arrangement (factory Cut) same as wall.

9. **False ceiling.** The false ceiling will be provided in the shelter (**including verandah**) both in first floor & ground floor. Details are as given below :-

(9.1) The false ceiling shall be made of 40mm thick PIR panels. The Panels will be made of 0.5mm thick PPGI on both side with 50mm thick layer of rigid CFC free polyurethane foam of density 40 kg +/- 2kg/ cum as Insulation. False ceiling panels are to be manufactured in single length & shall be installed covering the entire width of the living area resting on wall panels on either side. The false ceiling panels will have tongue & groove arrangement & shall be interlocked using cam locks systems. MS 'T' of 75 mm x 75 mm x 10 mm shall be provided to lay ceiling panels as per **Drg' DSB- 03** attached. Aluminum powder coated 'L' flashing of size 40 mm x 40 mm x 3 mm shall be used to fix all around the joining point of wall panel & false ceiling and vertically from floor level to the false ceiling level joining the walls at the corner junction for support & joining by means or rivets at minimum distance of 200 mm on both the faces.

10. **Doors**.

(10.1) **Door size.** 950 mm x 2100 mm.

(10.2) **Specifications of doors are as under :-**

(10.2.1) Frame shall be made from aluminum frame using 2 mm thick aluminum powder coated section confirming to IS 513 1986

(10.2.2) It will also have 04 numbers 100 mm steel hinges welded into the frame.

(10.2.3) Shutter shall be made out of PIR panel of 30 mm thick fixed with aluminum frame 2 mm thick single leaf with fixed arrangement. The main doors shall be provided with fly proof shutter on outside of the door in addition to shutter specified earlier. The galvanized stainless steel wire mesh shall be of stainless steel and average aperture 1.18 mm.

(10.2.4) Shutter frame shall be made of aluminum sheet of 105 mm x 60 mm powder coated.

(10.2.5) Fly proof shutter will be openings in the verandah whereas main shutter will be opening inside the shelter.

(10.2.6) Door shutter be fixed with 02 Nos aluminum aldrop of 300 mm length, aluminum tower bolt of size 200 mm at the top and 100 mm at the bottom on each shutter, one sliding bolt 300 mm long and one No 1500 mm long bow handles with fixing arrangement.

11. **Windows.**

(11.1) All windows shall be provisioned at sill height of 900mm. Window shall be of size 1000mm x 1000mm. Window shall be triple track sliding type made of **UPVC white / off white** colour (Make : Anglian, Everest, Duraflex, Quick slide) of size 63mm x 38mm x 2mm frame with 4mm thick glass pane with two shutter glazed and one shutter gauged third shutter shall be having stainless steel fly mesh of 0.56mm thickness and 1.4mm aperture complete in all as directed.

(11.2) Necessary locking arrangement shall be made for the window shutters.

(11.3) The wall panel housing the window to be precut to the window specification ex-factory and as per the spacing shown in the drawing attached.

12. **Ventilators.**

(12.1) Aluminum ventilators of size 1000 mm x 450 mm with two glass panels and fixed steel wire mesh on outer side to be provided. The ventilator frame will be manufactured using suitable Hindalco/ Jindal aluminum powder coated sections. The ventilator shall have double glass of 5mm thick for full length of 1000mm and width of size 300 mm, outside one fixed from top and inner side one fixed from bottom so that there is staggered gap in between them for air ventilation. The fly proof shall be of stainless steel wire mesh with 1.18 mm aperture fixed on outer frame work. The ventilator frame should be so designed that it gets firmly fixed with wall panel.

(12.2) Suitable arrangement shall be made to remove the glasses to enable cleaning of glasses and also the wire mesh

(12.3) The wall panel housing the ventilator to be precut to the ventilator specification ex factory

(12.4) The No of ventilators are same as of windows, fixed exactly above and in line with the windows, 300 mm from the false ceiling level.

13. **Sunshades**. Suitable sunshades at the slope of 20 degree downward will be provided for all external doors and windows not covered in verandah. Material used for making sun shade shall be pre coated pre profile steel sheet of 0.6 mm thick on angle iron frame 25 mm x 25 mm x 3 mm with fixing arrangement. Pre coated steel sheets shall match with the colour of the outer colour of wall panels. The minimum projection of the sunshade shall be 450 mm and 300 mm wider than the width of the openings.

14. **Finishes**. All steel work shall be given two coats of red oxide zinc chromite primer by the manufacturer in the factory. All exposed steel members shall be given two coats of colour paint as approved by the Commanding Officer, 116 Engineer Regiment.

15. **Foundation**. Foundation drawings of the structures and foundation bolts, nuts and all associated accessories shall be provided & constructed as per drg and direction by Engr-in-Charge.

16. **<u>Flooring</u>**.

(16.1) **<u>Ground floor</u>**.

(16.1.1) 150 mm thick sub base of hard core of broken stone aggregate.

(16.1.2) 100 mm thick base course in PCC 1:4:8.

(16.1.3) 100 mm thick layer of PCC 1:2:4, type BO over base course.

(16.1.4) 1200 mm x 600 mm x 10 mm thick non skid ceramic digital Vetrified tiles shall be provided over 20 mm thick screed bed in CM1:4.

(16.2) **<u>First Floor</u>**. Provision of material and flooring of the first floor shall be carried out by the firm. The specification for the flooring shall be as under :-

(16.2.1) MS decking plate of 1.0 mm thick.

(16.2.2) 100 mm thick reinforced concrete 1:11/2:3 over steel decking all as per design. The design to be vetted by NIT and approved by Commanding Officer.

(16.2.3) 600 mm x 600 mm x 10 mm thick non skid ceramic Vetrified tiles shall be provided over 20 mm thick screed bed in CM1:4.

17. **Dwarf Wall for Plinth**. Dwarf will be done as per drg and direction by Engr-in-Charge.

18. **Plinths**. Plinth will be done as per drg and direction by Engr-in-Charge.

19. **Foundation of Column.** Foundation for the column shall be constructed as per the drawings att and four Nos of foundation bolts 685 mm long 20 mm dia with nuts for each column will be provided.

20. **Eve Board**. Bird board of size minimum 30 cm width will be provided of pre coated plain sheet of 0.6 mm in Z section shape. Colour of Bird Board should be matching with the colour of the wall panel.

21. **Curtain Rods with Fixing Arrangements**. Curtain rod arrangements shall be provided on all doors and windows. The specification for the same is as under :-

(21.1) Mtrl - Mild steel powder colour coated fancy type drapery rod

(21.2) Size - 25 mm dia.

(21.3) Over hang projection - 450 mm on both sides.

(21.4) Fixing arrangement - Load of the rod to be on main columns.

(21.5) Brackets for holding the rods - Adequate number to support the rod.

22. **Gap Filling**. Minimum 15 Nos Silicon Tube for gap filing shall be provided with 2 x Silicon machine.

23. **Workmanship**.

(23.1) Welded connection will be provided conforming to IS-806-1968.

(23.2) Fabrication provisions in section 11 of IS-800 -1984 will be apply to all types of steel being used for fabrication.

(23.3) Various fasteners and fittings of mild steel shall be supplied unless otherwise Specified in and shall be fixed where required. Fastening means will be provided for the items to be fitted at the time of erection. 10% spare nuts, bolts, screw, hooks and washers etc will be provided per shelter. The fasteners supplied shall be conform to relevant BIS specifications.

24. **Electrification**. Electrical points for lights/fans/power shall be provided as given in the store list attached in **Annx–II**. In addition main board/MCB DBs comprising of SP/DB MCBs shall be provided. All electrical items (MCB/switches/Cables etc) shall be ISI marked and confirming to relevant IS specifications.

25. **Fan Hooks**. Hooks for fixing of ceiling fans are to be provided with supporting rod connected with the 'T' joint supporting the false ceiling panels with proper arrangements for securing it with help of nuts and bolts along with safety locking pin.

26. **Minor Details**. Details not mentioned in the specifications but required as per sound and good engineering practice shall be provided by the supplier. Only superior specifications shall be accepted, after obtaining due clearance by the Commanding Officer.

27. **Material**. All materials required for manufacture of the shelter shall be new and Materials shall comply with relevant bureau specifications and bear the mark.

28. **References of drawings.** Drawing att.

29. Various members and their quantities to be provided with each shelter as given drawings enclosed is tentative and an indicative only. The exact quantities to be provided with each shelter shall be as per actual requirement.

30. **Materials Make** :- The quality and sizes of the various types of items provided or used while supply or fabricating shall be conforming to relevant technical specifications. **The make that are offered will be intimated for approval of Accepting Officer within 07 days from date of placing of supply order.** Once make approved as offered and accepted will be final and all materials supplied under supply order will conform to the approved make offered. In case of any disputes / missing of details on account of specifications / makes, bidders are requested to intimate to tender inviting officer before bid quoting. Once tender is accepted no deviation shall be entertained. In this case decision of Accepting Officer shall be final and binding.

31. **Quality Control**. Accepting officer is free to get all parts/any part checked for quality conformation as given in the technical specifications irrespective of DGQA inspection. Cost of testing will be borne by the supplier.

NOTE: 1. The supplier will be whole and sole responsible to supply the items and materials required for construction as per specifications and good engineering practice including minor extras.

(Ajay Londhe)
Brig
Chief Engineer

Station : C/o 99 APO

Dated : 18 Jun 2026

Appx'B'
(Ref Ser No 01 of Costed Sch to TS No 1 of Job No OT/EC/3/56/2026-27/ 2653

TECHNICAL SPECIFICATION OF OR LIVING SHELTER D/S 18.0 X 6.10 M WITH 1.50 M WIDE VERANDAH

1. **Foundation**. PCC foundation will be made by using 400mm x 200mm x 200mm pre casted concrete blocks, in cement mortar 1:6 over 100 mm thick PCC 1:4:8 type D2 by using 40mm graded stone aggregate over 75mm thick 3" graded stone aggregate stone hard core all as shown in drg No 01 of 01 (PCC fdn details) att. 15mm thick cement plaster in CM 1:4 will be done on block masonry above ground level on the external surface.

2. **Column**. 18 Nos of columns, 10 No Columns of size 550mm x 330mm and 8 Nos of columns of size 500mm x 300mm by using RCC 1:2:4 type B1 using 20mm graded crushed stone aggregate. The size of footing – I & Footing – II will be 1500 x 1600mm and 1400 x 900mm over 75mm thick PCC 1:4:8 type D2 over 75mm thick hardcore over compacted soil. Complete work to be executed as per drg details att.

3. **Plinth Beam**. Plinth beam will be of size 300mm x 230mm using RCC 1:2:4 (20mm graded aggregate). 06 x 12mm TMT Bar be used for reinforcement and 8mm TMT bar will be used as stirrups at a distance of 200 c/c over 75mm thick PCC 1:4:8 40mm graded aggregate.

4. **Flooring**:-

(4.1) **Ground floor**.

(4.1.1) 150 mm thick sub base of hard core of broken stone aggregate.

(4.1.2) 100 mm thick base course in PCC 1:4:8.

(4.1.3) 100 mm thick layer of PCC 1:2:4, type BO over base course.

(4.1.4) 1200 mm x 600 mm x 10 mm thick non ceramic colour Digital tiles shall be provided over 20 mm thick screed bed in CM 1:4.

(4.2) **First Floor**. Provision of material and flooring of the first floor shall be carried out by the firm. The specification for the flooring shall be as under :-

(4.2.1) MS decking plate of 1.0 mm thick.

(4.2.2) 100 mm thick reinforced concrete 1:1 1/2:3 over steel decking all as per design. The design to be vetted by NIT and approved by Commanding Officer.

(4.2.3) 600 mm x 600 mm x 10 mm thick non skid ceramic tiles shall be provided over 20 mm thick screed bed in CM1:4..

5. **Plinth Protection**. All around the shelter, plinth protection of 750 mm wide, 75mm thick with PCC 1:3:6 type C1 shall be laid over 75mm thick hard core over compacted soil.

6. **Steps**. A step shall be constructed out of PCC sold brick masonry in CM 1:6 in front of the verandah in alignment with riser as 200mm and treads as 300mm.

7. Store list of constr mtrl att as **Annx 'I' & Annx-II**

8. **Materials Make** :- The quality and sizes of the various types of items provided or used while supply or fabricating shall be conforming to relevant technical specifications. **The make that are offered will be intimated for approval of Accepting Officer within 07 days from date of placing of supply order**. Once make approved as offered and accepted will be final and all materials supplied under supply order will conform to the approved make offered. In case of any disputes / missing of details on account of specifications / makes, bidders are requested to intimate to tender inviting officer before bid quoting. Once tender is accepted no deviation shall be entertained. In this case decision of Accepting Officer shall be final and binding.

9. **Quality Control**. Accepting officer is free to get all parts/any part checked for quality conformation as given in the technical specifications irrespective of DGQA inspection. Cost of testing will be borne by the supplier.

NOTE: 1. The supplier will be whole and sole responsible to supply the items and materials required for construction as per specifications and good engineering practice including minor extras.

(Ajay Londhe)
Brig
Chief Engineer

Station : C/o 99 APO

Dated : 18 Jun 2026

Appx 'C'
(Ref to Ser No 2 of CS to TS No 01 of
Job No OT/EC/3/2026-27/2653

DETAILED TECH SPECIFICATION FOR PATHWAY

1. **Earth compaction.**

(1.1) Forming embankments including raising or lowering of earth, spreading in layers, watering ramming/rolling.

(1.2) The compaction shall be done with the help of roller of 80 to 100 Kilo Newtons static weight with plain or pad foot drum or heavy pneumatic tyred roller of adequate capacity capable of achieving required compaction.

2. **Granular Sub Base (Soling).** Sub base shall be executed using GSB 150mm thick with stone aggregates/broken stone mixed with sand confirming to grading No 1 as per Table 400-2 of MORT&H and rolled with 8 to 10 Ton roller to required camber and gradient for **150 mm** spread thickness. The crushed or broken stone shall be hard durable and free from excess flat elongated soft and disintegrated particles dirt and other deleterious material.

3. **PCC 1:4:8.** **75mm** thick Plain Cement Concrete (PCC 1:4:8), Type D2, with using 40 mm graded stone aggregate as in sub base course. The crushed stone aggregate shall be hard durable and free from excess flat elongated soft and disintegrated particles dirt and other deleterious material.

3. **Sand Layer.** **40MM** Thick Sand Layer Cushion to be laid all over the track before laying of interlocking bricks. Sand should be free from excess flat elongated soft and disintegrated particles dirt and other deleterious material and should confirm to the grading of coarse sand. Test certificates in respect of bulking and moisture content should be submitted to this office before supply of material from approved Govt Lab and should be within permissible limits.

4. **Interlocking PCC Paver Blocks**

(4.1) Interlocking Paver Blocks **230 x 115 x 60 mm** minimum strength M35 Grade Concrete. Precast Block shall conform to IS-15658 2006. Solid Block made out of PCC 1:3:6 having the specified compressive strength of paver blocks at 28 Days should be 35 KN/Sqmm.

(4.2) The blocks are to be aid in herringbone pattern as attached in drawing.

(4.3) After laying of blocks the gaps is to be filled with fine sand and compaction of blocks is to be done by using plate vibrator.

(4.4) The shoulder of pathway to be laid with kerb stone size 300x300x100 mm with paint yellow and black .

6. **Additional Points.**

(6.1) Compressive strength of paver block as per IS code is to be submitted before laying of blocks.

(6.2) Complete all as specified in Drg's.

OTE:

1. The supplier will be whole and sole responsible to supply the items and materials required for construction as per specifications and good engineering practice including minor extras.

Station: c/o 99 APO

Dated : 18 Jun 2026

(Ajay Londhe)
Brig
Chief Engineer

Appx 'D'
(Ref Ser No 3 of CS to TS No 01 of
Job No OT/EC/3/56/2026-27/2653

TECHNICAL SPECIFICATIONS FOR SINGLE SLEEPING BUNK WITH COIR MATTRESS

1. **MS Rectangular Hollow Section**. Mild steel rectangular hollow section shall comply with IS-4923:1997 specification. Rectangular hollow section fitting shall be clearly finished straight and free from scale, crack, surface, lamination and other defects. MS Rectangular hollow section including fittings of size as shown in drawing used for manufacture shall be as per approved design and drawing to provide adequate strength and life of single sleeping bunk.

2. **Plywood Base.** Plywood base of size **1981.20 mm x 920mm x 12mm** thick of synthetic resin Bonded Grade Boiling Water Resistant (BWR) type, termite resistant and treated with preservative as per IS–5539-1969 shall be used as base over the iron frame for laying of mattress. The plywood will be fixed to the frame with flat head self tightening screws.

3. **Nuts and Bolts**. Hexagonal bolts and nuts will be provided as per design requirement conforming to IS : 1363-1967 shall be used for assembling of individual members. Allen key bolts/flat headed self tapping bolts shall also be used as required 10% extra quantity of all nuts, bolts and washers will be supplied to cater for losses and breakages in transits.

4. **Welding**. Proper welding joint will be formed where specified as per the design. All welding works will be as per IS: 823 – 1964 and IS : 816 – 1976. Welding shall be done by MIG welding process and shall be free from cracks and defects.

5. **Finishes**. The bed bunks shall be painted with two coats of anti-corrosive red oxide primer at the time of supply. Before application of paint all the steel surface shall be thoroughly cleaned to make the same free of rust, mill scale, dirt, oil etc either by mechanical or chemical means. The bed bunks should be painted with two coats of silver grey paint at factory or on site after assembling. Similarly steel locker will be painted with two coats of enamel paint, colour as specified by the consignee over a coat of anti-corrosive primer applied through spray painting.

6. **Coir Mattress**. Coir mattress of size **1981mm x 900mm x 75mm** thick comprising of 50mm thick high density coir with hessian border pasted with 25mm thick high density foam (pasting to be done with dendrite adhesive) Mattress to be stitched with good quality OG cotton fabric or as colour approved by Commanding Officer with double stitching on the margin/edge all as per sample approved. **Make: Sleep well/Kurl-on/Duroflex**

7. **Minor Details**. Details not mentioned in the specifications but required as per sound and good engineering practice should be provided by the supplier.

8. **Reference of Drawings**. Drawing No 1/1 of Single Sleeping Bunk.

(Ajay Londhe)
Brig
Chief Engineer

Station : C/o 99 APO

Dated : 18 Jun 2026

Appx 'E'

(Ref Ser No 3 of Costed Sch to TS No 01 of Job No OT/EC/3/56/2026-27/ 2653

TECHNICAL SPECIFICATIONS FOR BED SIDE LOCKER (SMALL)

1. **Bed Side Locker**. Bed side locker of size 900mm x 500mm x 1000mm made out of 20 gauge CR steel sheet conforming to IS : 513:1994 amdt 1 & 2. The locker will have two compartments with two separate doors on the front side and a partition on the inner side in the middle dividing the width of the locker into two equal parts. Left Compartment will have two shelves placed horizontally at equal distance and right compartment will have one 20mm dia steel rod 65mm below the top level for hangar arrangement. Vertical reinforcement of 800/400mm to be fitted inside on both the doors as well as on right side of the locker (Hanger Side).

2. **Painting**. Two coats of enamel paint dark OG colour/as indicated by consignee over a coat of anticorrosive primer applied through spray painting. Before application of paint all the steel surface shall be thoroughly cleaned to make the same free of rust, mill scale, dirt, oil etc either by mechanical or chemical means.

3. **Lock**. The locker shall have Hasp & staples as locking arrangement and two steel handles fitted on the front doors. Two tower bolts of 100mm to be fixed inside the left door on top & bottom for locking arrangement. For restricting the move doors inside, two stoppers shall be provided one on the top for full width of the locker and second at the middle of the bottom (1400mm) all as shown in the drg.

4. **Base**. The locker will be placed on 2 Nos, 100mm high hollow base made with M.S sheet channels.

5. **Minor Detail** Minor details not mentioned separately in the specifications but required as per sound and good engineering practice shall be adhered to.

6. **Reference of Drawing**. Drawing No 1/1 of Bed Side Locker.

Station : C/o 99 APO

Dated : 18 Jun 2026

(Ajay Londhe)
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Appx 'F'
(Ref Ser No 3 of Costed Sch to TS No 01 of
Job No OT/EC/3/56/2026-27/2653

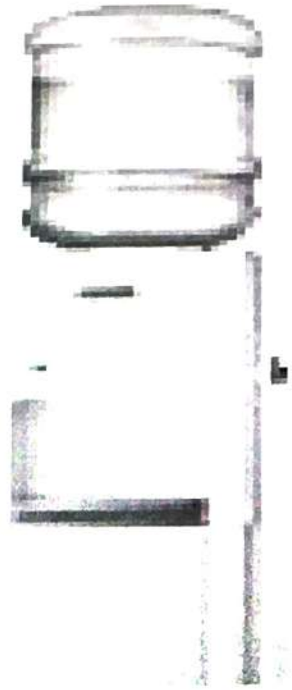
TECHNICAL SPECIFICATIONS FOR FURNITURE

1. **Chair Writing**. Chair writing of size **535mm x 470mm x 890mm** shall be provided. Chair shall be made out of well seasoned 1st class teak wood free from knots. All wooden exposed surfaces shall be finished with two coat of French polish high gloss finish after preparation of wooden surfaces. Plastic cane seat of size 470mm x 470mm x 535mm with arm rest of size 60mm x 25mm shall be provided. The legs will be fitted with hard PVC shoe. Refer Drg No FD 280 sheet 1/1.

2. **Writing Table**. Table of size **1145mm x 610mm x 760mm** shall be provided. Table top shall be 19 mm thick pre-laminated decorated particle board fixed with hollow section frame with screw @ 200 mm c/c. The frame work of table shall be of MS hollow section of size 25mm x 25mm all as per drg. End of legs will be sealed with MS sheet 1.6 mm thick with proper welding process to avoid the dust entry. 6 mm thick teak wood lapping shall be provided on all the exposed edges of MDF duly treated with French polish as specified. All steel surfaces will have two coats of synthetic enamel paint over a coat of red oxide primer. The legs will be fitted with hard PVC shoe. Refer Drg No WT 1/1.

3. **Bottled Water Dispenser**. Bottled Water dispenser body made of ABS plastic and tank made of Food Grade stainless steel. Capacity of cooling and hot water shall be of 3 liters and 5 liters respectively with 3 temperature taps like Hot, Plain and cold. Heavy duty bottle piercer with bottle holder shall be provided to hold the water bottle of capacity 20 ltrs. Bottom of the body shall consist of a refrigerator of 14 ltrs capacity. Make: Blue star / Usha.

WATER DISPENSER



4.0 **Powder based Fire Extinguisher (Red) - 1** : Dry powder fire extinguisher of 1 Kg Capacity in confirmation with ISO/CE certified with wall mounting holder. It shall be helium leak tested and with 3 years warranty. Make : Ceasefire / Suprema / Eco fire

(Ajay Londhe)
Brig
Chief Engineer

Station : C/o 99 APO

Dated : 18 Jun 2026

Annx 'I'
(Ref Ser No 01 of CS to TS No 01 of
Job No OT/EC/3/26-27/ 2653

STORE LIST OF CONSTR MTRL FOR OR LIVING SHELTER (D/S)

Ser No	Description of Stores	A/U	Qty	Remarks
1	Shelter Part only	Set	1	
Construction Material				
1	Cement :Portland pozzolana cement : PPC Cement IS-1489 (part-2) for clained clay based PPC and IS 1489 (part-1) for fly ash based PPC) packed in 50 Kgs HDPE bag conforming to IS 11652/2000 **Make : Ambuja/ ACC/ Ultra tech/ Dalmia/ Ramco/ Birla/ CCI)**	Bags	615	
2	**Coarse Sand** : Sand shall be clean, sharp, angular hard and durable free from adherent coating and shall not contain clay, mica and other impurities. The maximum quantities of clay and fine silt in sand shall not be more than 5% by mass as per IS-383-1970.	Cum	45	
3	**20mm Stone Aggregates** : Stone aggregate shall be crushed stone 20mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	38	
4	**40mm Stone Aggregates** : Stone aggregate shall be crushed stone 40mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	22	
5	**Hardcore 63 mm Stone Aggregates** : Stone aggregate shall be crushed stone not exceeding 63mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	24	
6	PCC Solid block 400x200x200mm (+/-10mm) minimum strength 50Kg/Sqcm. PCC shall conform to IS-2185 (Part I) 2005. Solid Block made out of PCC 1:3:6 having the compressive strength 5 KN/Sqmm.	Nos	632	
7	Non skid ceramic Coloured Digital tile of size **1200mm x 600mm** x 9-10 mm thick (Make:Kajaria/Asian/Orient/Johnson)	Nos	407	
8	White Cement (Make- Birla Gold)	Kg	50	
9	TMT Bar 8mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	330	
10	TMT Bar 10mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	1740	
11	TMT Bar 12mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	350	
12	TMT Bar 16mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	1030	
13	ACP(Aliuinium Composite Panal) Size 1.50*1.00	Nos	1	
14	Stainless steel 300 mm long Word	Nos	10	
15	Water proofing compound (Make- Dr. Fixit)	Kgs	8	
16	Lime for white wash	Kgs	15	
17	Binding Wire 0.90mm	Kg	20	
18	Ply wood 12mm thick size 8' x 4'	Nos	12	
19	Balli 100mm dia 3 mtr	Nos	40	
20	Wooden runners 2" x 2"	RM	250	
21	Nails	Kg	20	

Earthing Items				
22	LC set with safe chemical 25.1 Zinc & Alloy Coated Safe Earthing Electrode (76.3mm dia) 2 mtr long with back frilled compound conductotile - 2 Bags CPRI approved 25.2 Air Terminal single pointed aluminium rod 12mm dia and 600mm. 25.3 GI pipe med cgraded 12 mtr loing 40mm dia. Make TATA/JINDAL/BANSAL 25.4 Insulator with screw 20 Nos 25.5 Aluminium cable 70Sqmm multi standed single core-30 RM . Make PLAZA/HAVELLS/ RALLISON ISI Marked 10 RM. 25.6 Testing point terminal box made out of gun metal of phosphorus of size 75mmx75mmx5mm with screw. 25.7 Nut and bolts 2" x 1/2" - 08 Nos 25.8 Foundation Nuts & bolts 16mm x 400mm - 04 Nos. 25.9 Aluminium lugs 70Sqmm for connection of cable with nut & bolts of suitable size - 02 Nos. 25.10 Earthing pit cover- 01 Nos.	Nos	1	
23	Electric Earthing Set complete with items as given below :- 23.1 GI Earthing plate 600mm x 600mm x 6mm thick with two Nos holes - 01 No. 23.2 GI pipe 20mm light grade for watering - 1.2 Mtr. Make : Tata/Jindal/Prakash 23.3 GI wire 4mm dia - 02 Kgs. 23.4 GI funnel with wire mesh suitable for 20mm dia GI pipe - 01 No 23.5 GI strip (32mm x 6mm x 4.00mtr) 23.6 Common salt - 50 Kgs. 23.7 FRP earthing pit with cover - 01 No. 23.8 GI nut bolt with washer of size 10mm dia 25mm long - 2 Nos. 23.9 Charcoal - 50 Kgs.	Nos	1	

(Ajay Londhe)
Brig
Chief Engineer

Station : C/O 99 APO

Dated : 18 Jun 2026

Annx-II
(Ref Ser No 1 of CS to TS No- 1 of Job No OT / EC/3/56/2026-27/ 2653

STORE LIST OF ELECT ITEMS FOR 01 X LIVING SHELTER (ORL) (D/S)

Ser No	Description of items	A/U	Qty	Remarks
1	Surface and wall mounted LED tube rod 25/20 watt, 1600 lumens output with suitable fitting frame. Frame shall be powder coated with aesthetically designed dual colour ends caps with bipin holder, connector block complete. (Make: - Bajaj / Havells / Philips/Syska/Wipro/ Crompton.	Nos	30	
2	Pixel bulk head fitting in deep drawn anodized aluminium body, epoxy white powder coated and heat resistent prismatic glass cover with specially designed gasket with LED lamp 1X 15 watt. (Make -Bajaj / Havells /Phillips/Crompton.	Nos	4	
3	Surface and wall mounted LED batten with Niano diffuser technology to ensure minimum loss in light transmitted 8 watt, LED fitting including 9 watt tube rod complete with suitable fitting frame complete. (Make :-Cat No BN 550W LED 6-6500 (KLED 1 Ft) of Philips or equivalent in Crompton / Bajaj /Havells	Nos	4	
4	Ceiling Fan 1200 mm sweep with 5 step fan regulator one modular, conforms to IS:2312, ISI marked with double ball bearing with modular fan regulator. Fan hooks with MS clamps, bolts and nuts. (Make:- Fan - Khaitan/Usha/ Bajaj/Crompton Greaves. Modular Fan Regulator : Havells / Legrand/ Polycab / Anchor)	Nos	16	
5	Exhaust fan 300mm sweep, with heavy speed motor and louvers. (Make: – Khaitan / Usha/Crompton Greaves /Bajaj.)	Nos	6	
6	PVC insulated heavy duty 1100 volt grade cable with aluminium conductor 3.5 core and 16 sqmm served with sheathing of PVC tape and overall PVC sheathed, flat or circular. (Make:- Plaza / Paragon/ Polycab/ Usha/ Bajaj / Finolex / Havells	RM	100	
7	PVC sheathed unarmoured stranded copper conductor cable of size 6.0 sqmm dia conform to IS 694/1990 (green in colour of 90 Mtr Roll). (Make: - Plaza / Paragon /Finolex / Havells/Richa)	Roll	1	
8	PVC sheathed unarmoured stranded copper conductor cable of size 4.0 sqmm dia conform to IS 694/1990 (red, black and Green of 90 Mtr Roll)(Make: - Plaza / Paragon /Finolex / Havells/Richa)	Roll	1	
9	PVC sheathed unarmoured stranded copper conductor cable of size 1.5 sqmm dia conform to IS 694/1990 (red/ black in colour of 90 Mtr Roll).(Make: - Plaza / Paragon /Finolex / Havells/Richa)	Roll	10	
10	PVC casing capping 32mm wide, not less than 2 mm thick. (Make: - Polycab/Suncab/Indoplast/ Surya/ Anchor/ Prestoplast .)	RM	100	
11	PVC Junction linker, 75mm x 75 mm (Make : Prestotech/ Polycab/ Suncab/ Indoplast/ Payal/ AKG)	Nos	50	
12	Modular switch piano flush type 6 Amps 240 volts single pole one way. (Make :- Havells/ Henly /Anchor/ Legrand.)	Nos	62	
13	Modular socket outlet multipurpose 5 pin 6 amps flush button type. (Make : Havells /Anchor/ Legrand .)	Nos	30	

Ser No	Description of items	A/U	Qty	Remarks
14	Modular socket Combination multipurpose 6 Pin. 6 to 16 Amps.(Make :- Havells /Anchor/ Legrand)	Nos	4	
15	PVC switch board 8" x10" Conform to IS 14772. Make - Prestoteak/Polycab/Suncab/Indoplast.	Nos	15	
16	PVC switch board 7" x 4" Conform to IS 14772. Make - Prestoteak/Polycab/Suncab/Indoplast.	Nos	8	
17	PVC switch board 4" x 4" Conform to IS 14772. Make - Prestoteak/Polycab/Suncab/Indoplast.	Nos	30	
18	Fan regulator (Make:- Havells/Anchor.)	Nos	16	
19	Ceiling rose, surface bakelite, 65 x 50 mm, 2 to 3 terminals. (Make: - Anchor/Crabtree/ Legrand/ Havells.)	Nos	52	
20	House service bracket (Hook type) complete with 3 m long GI pipe 40 mm dia medium grade, 2 Nos 90 degree bends, MS hooks/clamps and stay or guy.	Nos	1	
21	MCB DB 4 Way SPN, 240 volt conform to IS 13032, (Make: Anchor/ Legrand/ Havells/IndoAsian/Polycab)	Nos	2	
22	MCB DB 8 way SPN. (Make:Anchor/Legrand/ Havells/ IndoAsian/ Polycab)	Nos	2	
23	MCB TPN 63 Amps. (Make: Anchor/Legrand/ Havells/ IndoAsian/ Polycab.)	Nos	2	
24	MCB SPN 40 Amps. (Make: Anchor/Legrand/Havells/Polycab/ IndoAsian)	Nos	4	
25	Insulation tape 1.80 cm width and 7.5 m long (Make: Anchor)	Nos	25	
26	No 8 x 19 CSK screw full threaded	Pkt	5	
27	No 8 x 32 CSK screw full threaded	Pkt	5	
28	No 8 x 50 CSK screw full threaded	Pkt	5	
29	No 8 x 75 Bolts and Nuts full threaded	Nos	10	

Station : C/o 99 APO

Dated : 16 Jun 2026

(Ajay Londhe)
Brig
Chief Engineer

Annx - 'III'
(Ref Ser No 2 of CS to TS No 01 of Job No-OT/EC/3/2026-27/ 2653

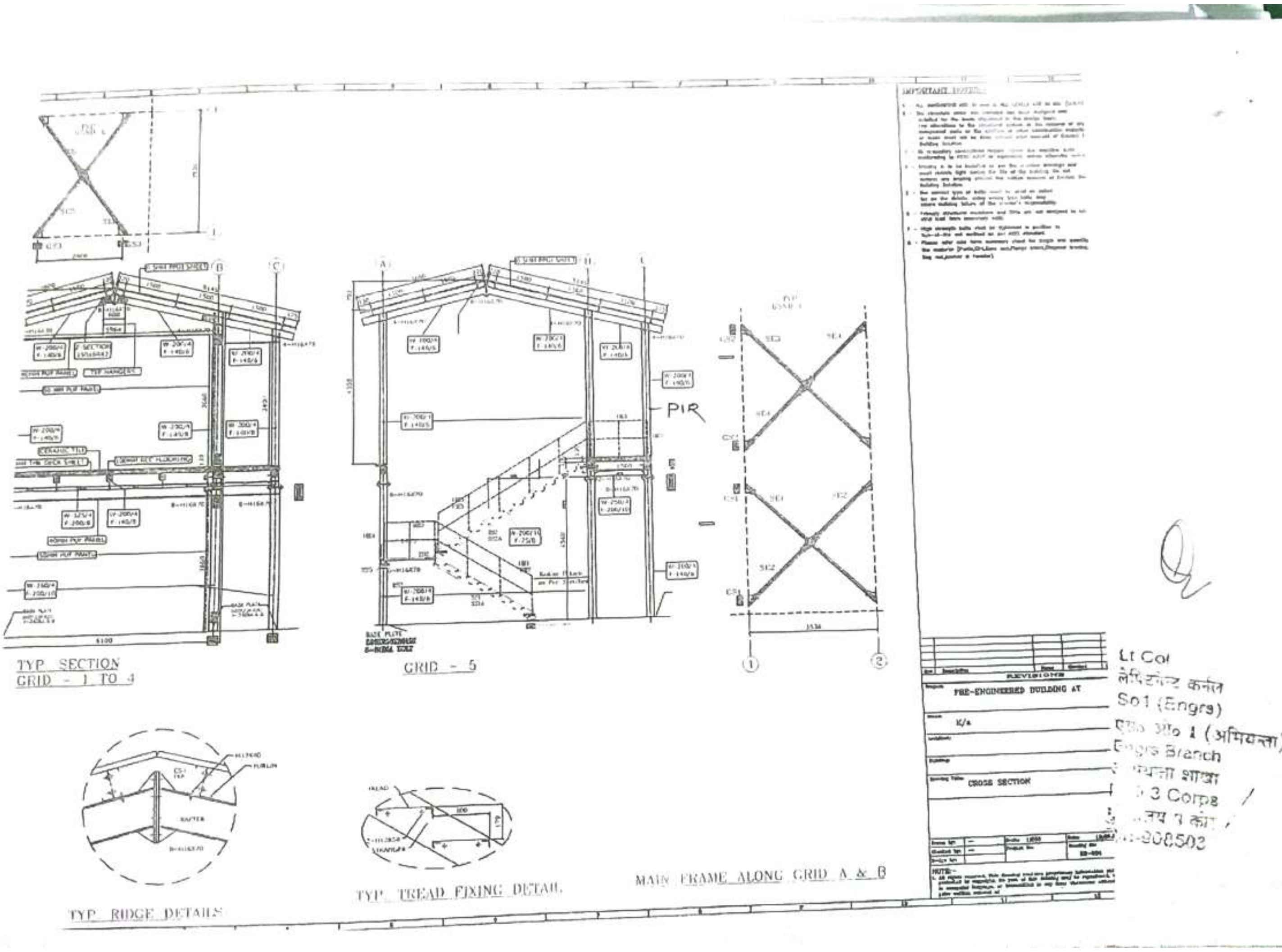
STORE LIST OF CONSTR MTRL FOR PATHWAY (AREA - 12.00 SQM)

Ser No	Description of items	A/U	Qty	Remarks
Construction material :-				
1	Cement :Portland pozzolana cement : PPC Cement IS-1489 (part-2) for clained clay based PPC and IS 1489 (part-1) for fly ash based PPC) packed in 50 Kgs HDPE bag conforming to IS 11652/2000 **Make : Ambuja/ ACC/ Ultra tech/ Dalmia/ Ramco/ Birla/ CCI)**	Bags	5	
2	**Coarse Sand** : Sand shall be clean, sharp, angular hard and durable free from adherent coating and shall not contain clay, mica and other impurities. The maximum quantities of clay and fine silt in sand shall not be more than 5% by mass as per IS-383-1970.	Cum	2	
3	**40mm Stone Aggregates** : Stone aggregate shall be crushed stone 40mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	1	
4	**Hardcore 63 mm Stone Aggregates** : Stone aggregate shall be crushed stone not exceeding 63mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	3	
5	Machine pressed precast concrete interlocking paver block of size 225 x 112.50mm confirming to IS 15658-2006 of 60 mm thickness M-35 Grade. And filling the joints with grey cement and coloured pigments as per required.	Nos	479	
6	Providing and laying precast PCC Kerb stone type B1 1:2:4 (20mm graded stone aggregate) of size 300mm x 100mm x 300mm jointing with cement mortar in Cm 1:3.	Nos	70	
7	Kerb stone Synthetic Enamel Paint (Yellow and black Colour)	Ltr	6	
8	Red oxide Primer	Ltr	2	
9	Turpentine Oil (Thinner)	Ltr	2	
10	Painting brush 3" of best quality.	Nos	1	

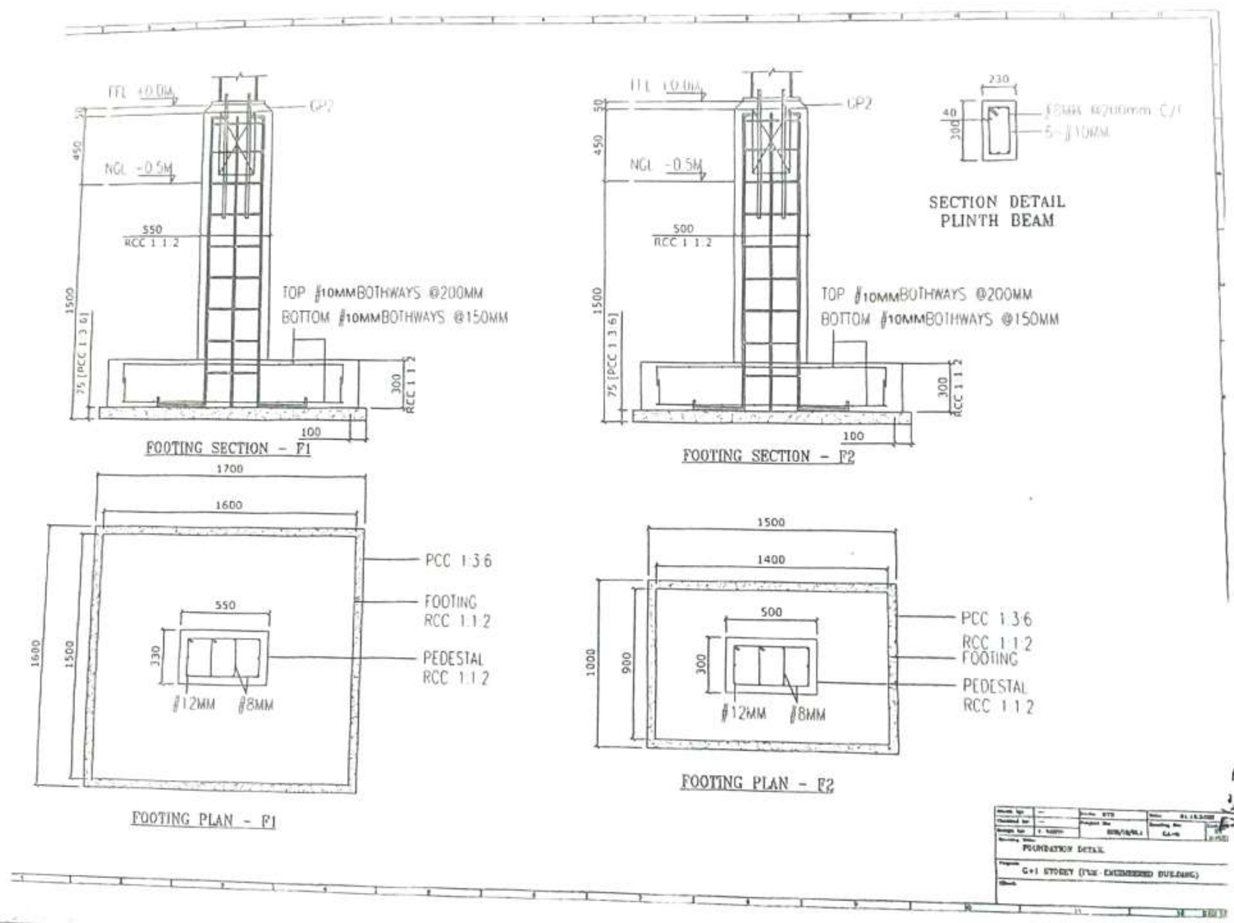
Station : c/o 99 APO

Dated : 1 6 Jun 2026

(Ajay Londhe)
Brig
Chief Engineer



Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
3 Corps



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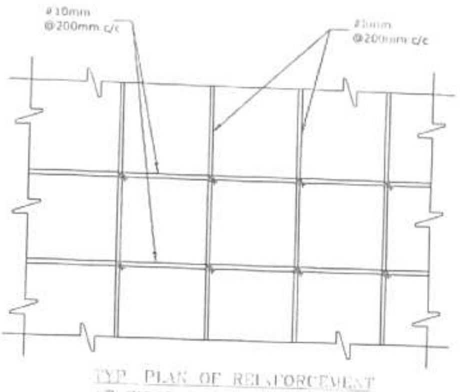
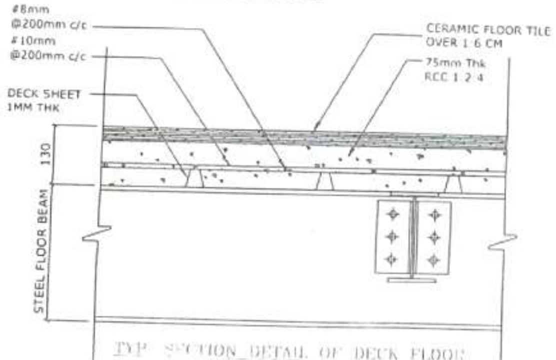


TYP. PLAN OF REINFORCEMENT
AT FIRST FLOOR

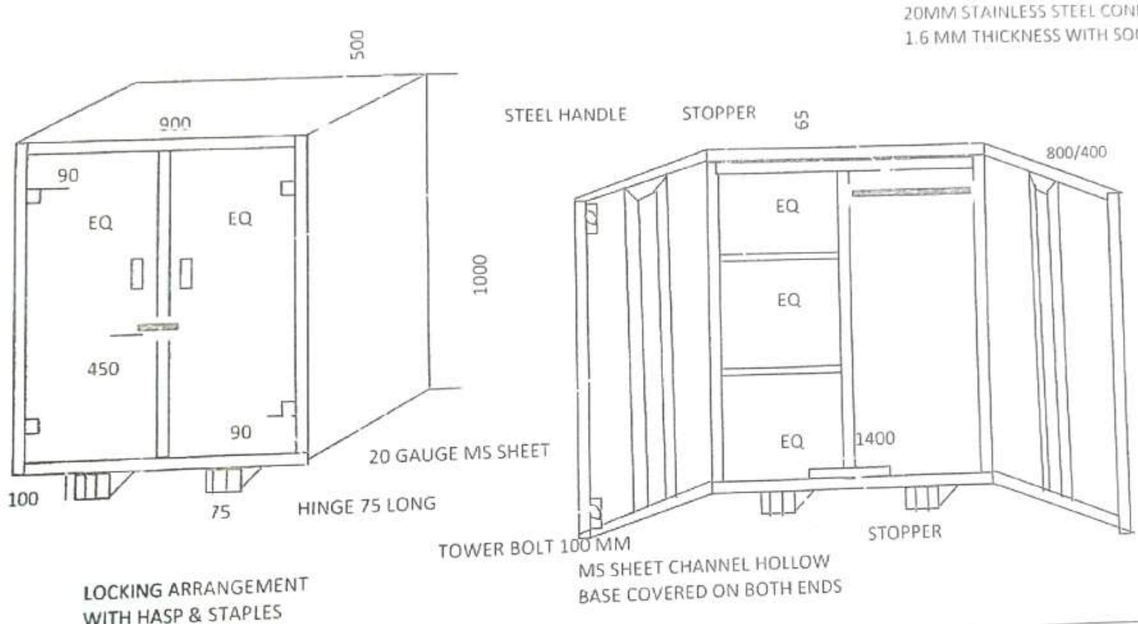


TYP. SECTION DETAIL OF DECK FLOOR
AT FIRST FLOOR

PEDESTAL & SLAB DETAIL

G+1 STOREY (PRE-ENGINEERED BUILDING)

BED SIDE LOCKER

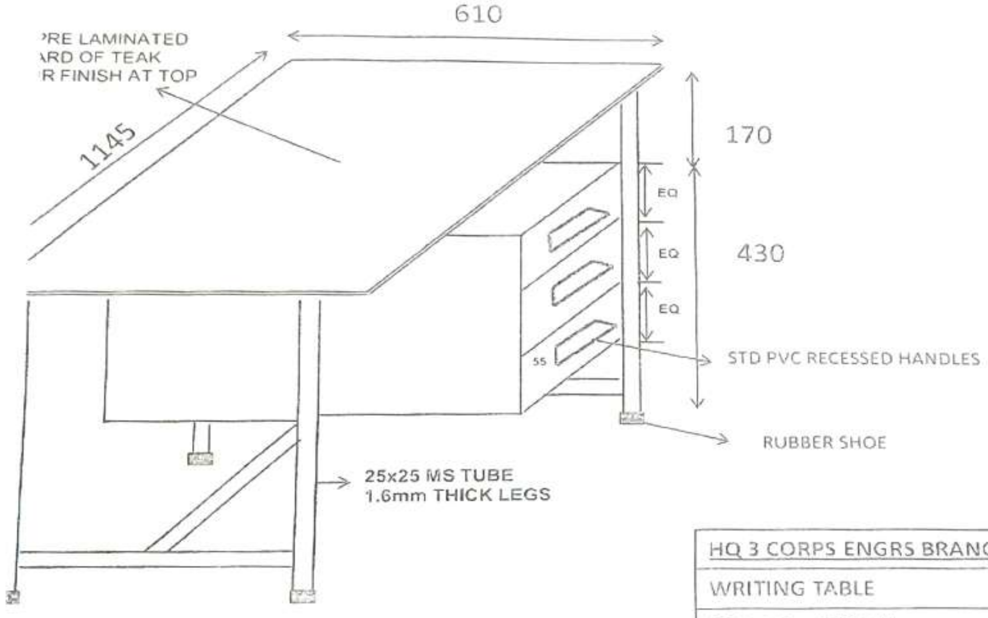


NOTE : ALL DIMENSIONS ARE IN MM
AND DRAWING NOT TO SCALE

H Q 3 CORPS, ENGRS BRANCH
BED SIDE LOCKER
DRG NO : BSL 1/1
GSO 1 (OP WKS)

Lt Col
लेफ्टिनेंट कर्नल
SO 1 (Engrs)
(अभियन्ता)
Engrs Branch
3 Corps

WRITING TABLE



DIMENSIONS ARE IN MM AND NOT TO SCALE

HQ 3 CORPS ENGRS BRANCH
WRITING TABLE
DRG NO : WT 1/1
GSO1 (OP WKS)

Lt Col
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So1 (Engrs)
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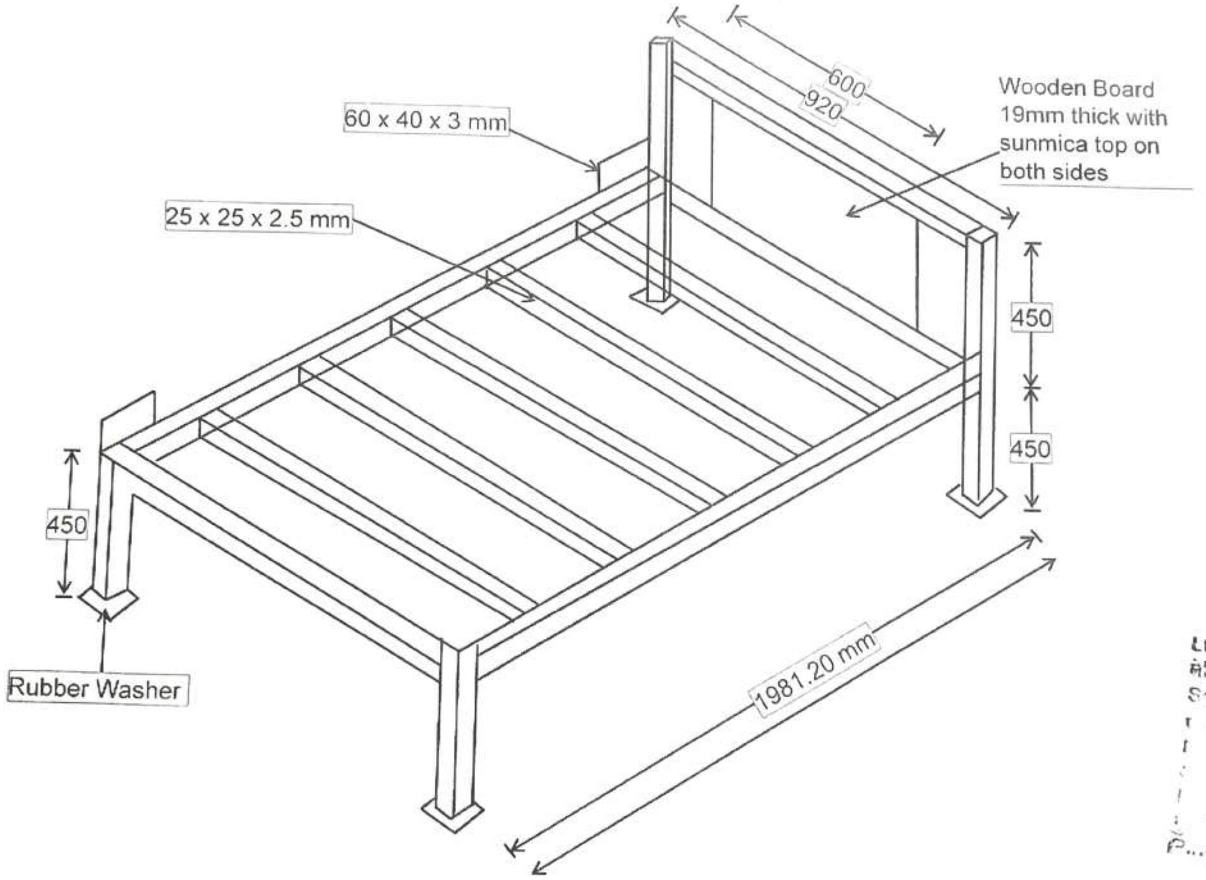
SINGLE SLEEPING BUNK



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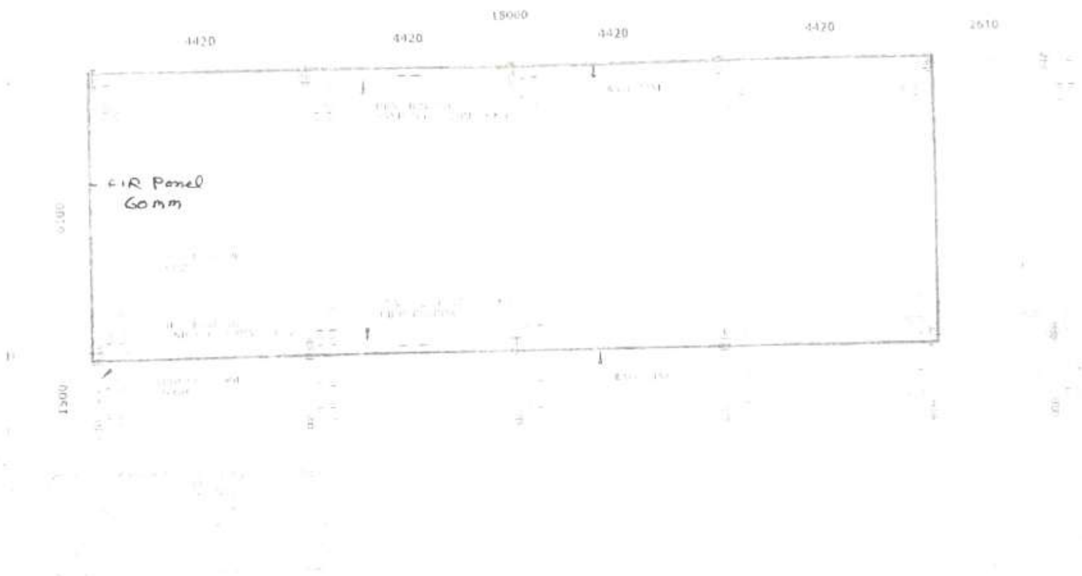

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Engrs Branch
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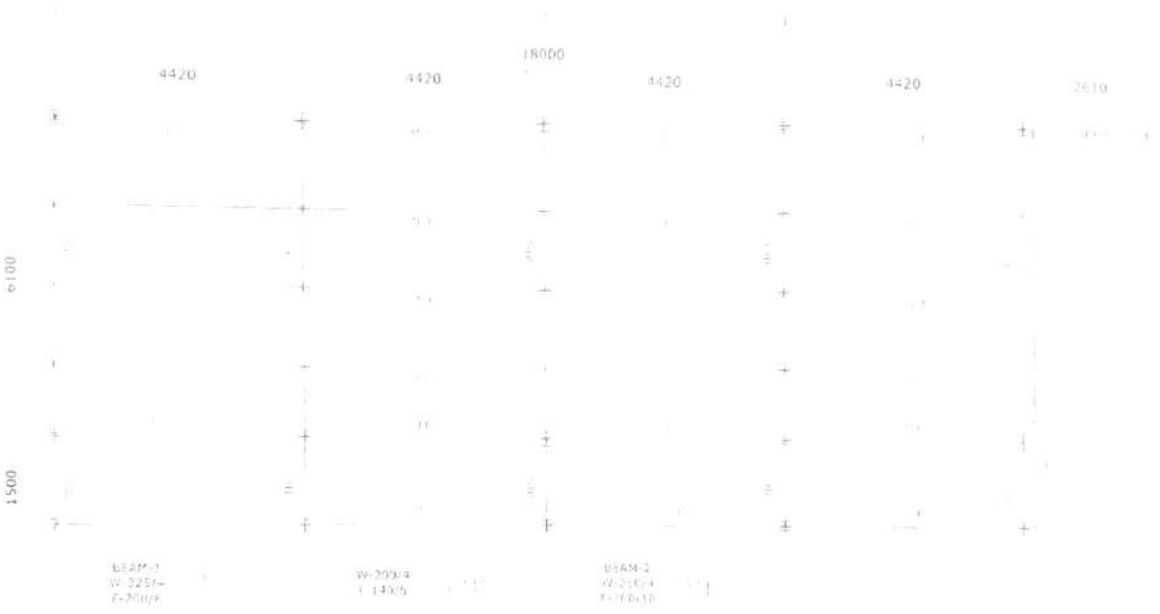
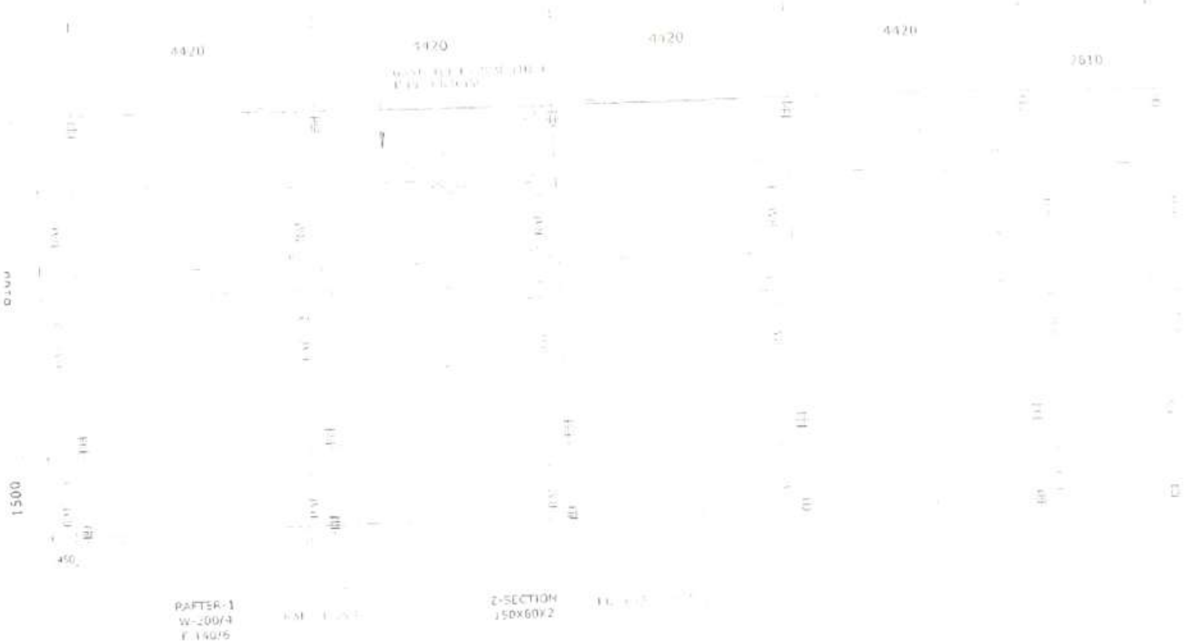


Lt Col
Corps
Pin-908503



Prakash Vaidya

Lt Col
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So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
HQ 3 Corps
मुख्यालय 3 कोर
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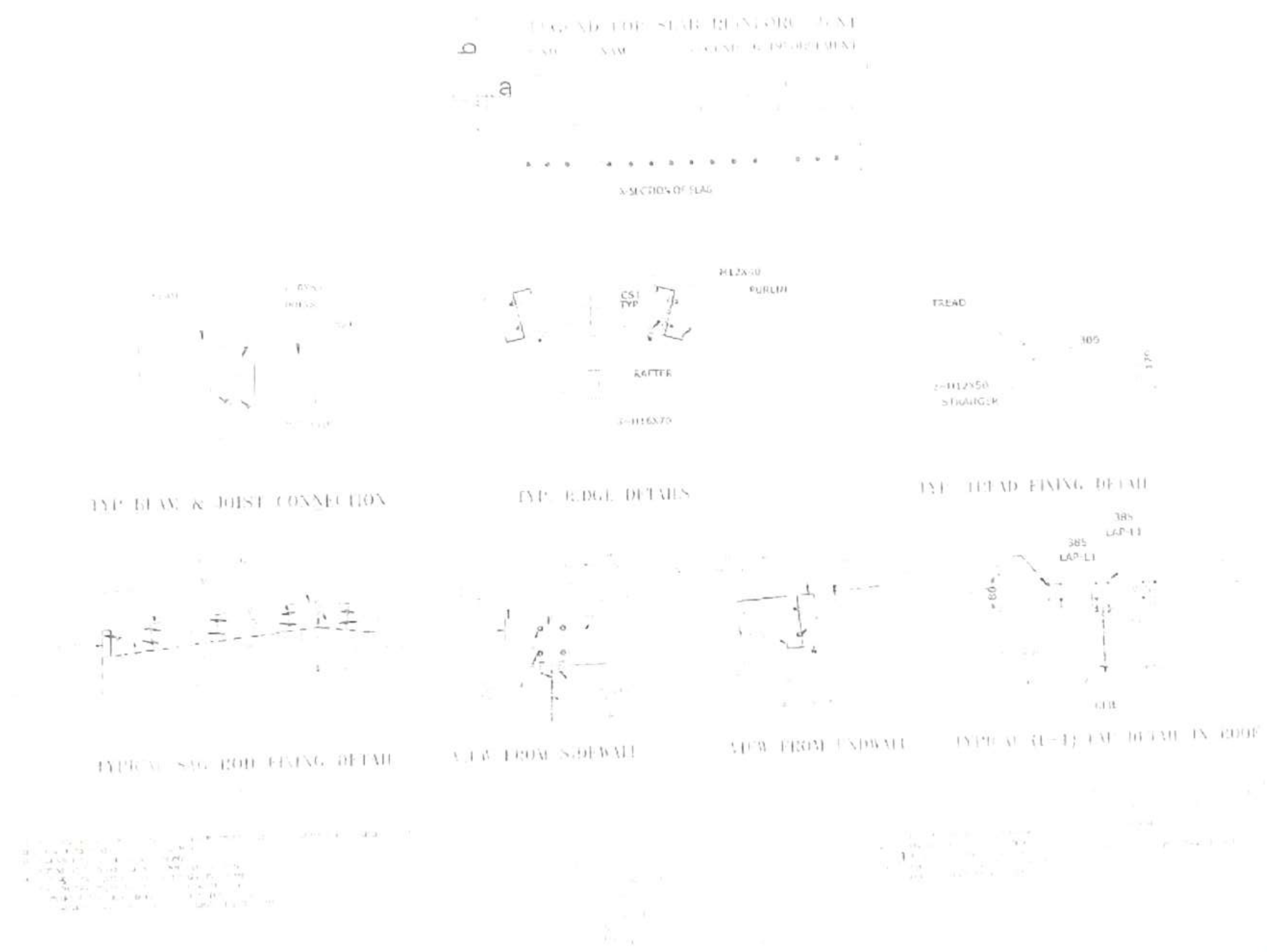

Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
3 Corps
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COSTED SCHEDULE OF WORK

Name of Work : Provn of qty 02 x Living Shelter (ORL) D/S at 56 Inf Div AoR

Job No : OT/EC/3/56/2026-27/2655

Ser No	Description of work	Brief Specifications		A/U	Qty
1	Provn of qty 02 x Living Shelter (ORL) (D/S)**(Shelter Part , Constr Mtrls including E/M Items)**	Provn of qty 02 x of Pre-Engineered Portable Frame OR Living (Double Storey) PIR Shelter of size 18.0m x 6.1m with 1.5m wide verandah including roof covered steel staircase with SS handrails on one side of the shelter as in drawing. The structure made out of primary members of High Tensils Steel (340 Mpa min) and Secondary members of (240 Mpa) steel. The main wall and portion and partition wall if any, will be made up of **60mm thick** PIR Panels. The Mezzainine Floor shall be with 1.0 mm Thick Galvanized Deck sheet. Layout of partition wall of ground and first floors. if any, will be as per the respective floor plans attached. The main beam of stair case are made up of built up C section of size 75mm, 200mm, 8/10mm and step platform are mae up of chequered plate 4mm thick. The side railings are made up of square hollow steel sections of 32mm x 32mm x 2.9mm @ spacing not greater than 300mm. The roofing will be made of 0.5mm PPGI colour coated sheets. Detailed specification of shelter and constr part as given in **Appx 'A'** and **Appx 'B'** att. Store List of elect items and constr mtrl are att as per **Annx - I &** II.		Set	2.00
2	Provn of qty 02 x Living Shelter (ORL) (D/s) **(Pathway) 2 x (10.00 x 1.20 mtr)**	Provn of interlocking paving blocks (size 10.00x 1.20 mtr) for pathway as per detailed specifications given as per **Appx 'C'** and store list as per **Annx - III.**		Job	2.00
3	Provn of qty 02x Living Shelter (ORL) **(Furniture Part)**	5.1	Provn of **Single Sleeping Bunk with Coir Mattress** as per tech specification given in **Appx 'D'**	Nos	80.00
		5.2	Provn of **Bed Side Locker (Small)** as per tech specification given in **Appx 'E'**	Nos	80.00
		5.3	Provn of **Chair Writing Wooden** as per tech specification given in **Appx 'F'**	Nos	8.00
		5.4	Provn of **Writing Table** as per tech specification given in **Appx 'F'**	Nos	8.00
		5.5	Provn of **water Dispencer** all as per tech specifications given in **Appx 'F'**	Nos	4.00
		5.6	Provn of Powder based **fire extinguisher** (Red) - 1 as per tech specification given in **Appx 'F'**.	Nos	8.00

Appx 'A'
(Ref Ser No 1 of CS to TS No -1 of
Job No OT/EC/3/56/26-27/2655

DETAILED TECHNICAL SPECIFICATION OF OR LIVING SHELTER D/S OF SIZE (18 M X 6.1M WITH 1.5 M WIDE VERANDAH) : SHELTER PART

1. Design, supply & erection of Pre-Engineered portal frame **Double Storey PIR Shelter** of size 18.0 m x 6.1 m with 1.5 m wide verandah including roof covered steel staircase with SS handrails on one side of the shelter as in drawing. The structure made of primary members of High Tensile Steel (340 Mpa min) Yield Strength and Secondary members of (240 Mpa) Yield Strength of steel. The main wall and partition wall if any, will be made up of **60** mm thick PIR Panels. The Mezzanine Floor shall be with 1.0 mm Thick Galvanized Deck sheet as per specification attached. Layout of partition walls of ground and first floors, if any, will be as per the respective floor plans attached. The main beam of stair case are made up of built up C section of size 75 mm, 200 mm, 8/10 mm and step platform are made up of chequered plate 4 mm thick. The side railings are made up of square hollow steel sections of 32 mm x 32 mm x 2.9 mm @ spacing not greater than 300 mm. The roofing will be made of 0.5mm PPGI colour coated sheets. Other specifications are as per description given under each specification and drawings Nos DSB - 01 to 03 attached.

2. **Design Parameters.**

(2.1)	Seismic Co-efficient	-	As per Seismic Zone VI.
(2.2)	Wind Load	-	Equivalent to wind speed of 55m/sec as per IS-875.
(2.3)	Roof Slope	-	1:3
(2.4)	External Temperature	-	(-)10°C to (+)50°C
(2.5)	Water Absorption/ Penetration	-	Nil
(2.6)	Termite Proofing	-	Termite Proof.
(2.7)	Fire Resistant	-	Fire retardant and should not emit toxic fumes.
(2.8)	Ease of Constr	-	Shelter should be modular in design.
(2.9)	Design Load	-	As per IS- 875.

3. **Dimensions.**

(3.1)	Length	-	18.00 Mtr.
(3.2)	Width	-	6.10 Mtr.
(3.3)	Verandah width	-	1.50 Mtr.
(3.4)	Internal height of each Storey upto false ceiling	-	3.66 Mtr.

4. **Details of Structural Members.** The steel structure for the shelters shall be designed for the design parameters as enumerated in Para 2 and for the dimensions indicated in Para 3 and drawings attached. The detailed design calculations, structural drawings and foundation design shall be submitted by the firm duly vetted by chartered structural Engineer/IIT/NIT.

(4.1) Roof members and columns shall be designed to stand max wind speed of 55 m/sec. The steel should be conforming to IS-1161of 1979 or IS 2062-1984.

(4.2) The roof shall be provided with 0.50 mm colour coated trapezoidal PPGI sheet.

(4.3) The roof shall have minimum projection of 0 45 Mtr from the eaves wall. Purlins shall also be provided in the roof at the eaves and at gable roof extension (outside the structure)

(4.4) **Steel Stair Case.** The main beam of stair case made up of built up C section of size 75 mm, 200 mm, 8/10 mm and step platform are made up of chequered plate 4 mm thick. The side railing are made up of square hollow stainless steel sections of 32 mm x 32 mm x 2.9 mm @

spacing not > 300 m. The staircases would be provided overhead roof in continuation with the shelter roof and roof projection of 0.45 m on gable side is also provided. The stair case is opened on all other three sides.

(4.5) **Foundation bolts**. Foundation bolts of size dia 20 mm x 685 mm long. Four numbers for each foundation. The bolts must be made from hard steel.

5. **Roofing**.

(5.1) **Roof**. The roof slope shall be laid in 1:4. Roofing shall be done with roof sheet of PPGI 0.50 mm with guard film. The PPGI sheet shall have minimum coating of 4-5 micron epoxy primer and 18-20 micron polyester top coat. The top of 0.50 mm PPGI will have a guard film of 30-40 micron on the finished surface only for protection against scratches. The Pre coated GI sheet will confirm to IS-14246-1995. Base metal of PPGI will be of CRCA as per IS-513 and galvanised as per IS-277 having zinc coating mass of 120 GSM. The roof shall be laid over a frame work of trusses, purlins and column fixed using tapping screw with steel washer.

(5.2) **Ridge Cover**. Ridge cover of size 225 mm x 225 mm x 0.5 mm made from pre coated plain sheet matching with the roof sheet shall be provided.

(5.3) **Roof projections**. The roof shall have minimum projections 0.45 m for the eaves wall in the rear in front where no verandah is provided and also on both gable walls. Purlins will be provided for the roof projections at the eaves and at gable ends (outside the structure made from MS pipe 80 NB (H) conforming toIS-1161-1779).

(5.4) **Overlapping**. Sheets to have minimum overlap of 150 mm.

6.0 **Wall Panels (Internal/External)**- All walls shall be made of PIR insulted panel with insulation core polyisocyanurate foam (PIR) insulation. **PIR insulated panel shall be 60 mm thick** with **density 40 Kg/Cum or more**. The panel insulation material in the panel shall have fire retarding and self -extinguishing properties as per international standard. There should not be any gap between wall panels and wall and roof by provision of suitable joining arrangement. Other details of wall panels are as follows.

6.1 All material required for the manufacture of shelter shall be new and shall comply with relevant Bureau of Indian Standard Specification.

6.2 The bulk density of PIR panel insulation should be **40 Kg /Cum or more insulation core (PIR) polyisocyanurate foam** and the dispensing machinery should be kept with a PLC controlled panel for monitoring and controlling the injection rate to assure specified uniform density requirements. The total thickness of the finished composite panel should be 51 mm. The tolerance in the panel can only be on the plus side

6.3 The outer and inner colour coated (RAL-1001) HI skin made from hot dipped galvanized steel of the panels should be 0.50 mm thick TCT (Total coated thickness). Each panel should be supplied pre-painted with a colour coating of 50 microns of architectural polyester on a minimum 120 GSM base of Zink coating on the finished surface only for protection against scratches during handling and transportation. Base metal of GI skin CRCA should be as per **IS-513-2016 (reaffirmed 2021)** and grade of zinc coating will be 120 GSM as per IS-277-2018 **(reaffirmed 2022)**

6.4 The pre-coated GI sheet skin should have minimum coating of 4-5 micron epoxy primer and 25 micron polyester top coat on the finished surface and 7-8 micron primer Alkyd back coat which is bounded to the (PIR) polyisocyanurate foam. The pre coated GI sheet should conform to **IS-14246-2013 (reaffirmed 2018)** with manufacture test certificates conforming to specification.

6.5 The PIR insulated core of these composite panels shall have the following properties :-

6.5.1 Density (IS 11239: Part 2) - **40 Kg/ m^3 or more**

6.5.2 Panel wt should not be exceed - Max 11.70 Kg/ m^3

6.5.3 U value of PIR panel will not exceed – 0.30 w/m2K

6.5.4 Flammability - BS2D0(min)

6.5.5 Blowing Agent – Non CFC & Non HCFC

6.5.6 Compressive Strength at 10 % deformation - 115 KN/m^2

6.5.7 Tensile Strength (Adhesion Strength) (IS 16203) – 2.5 Kg/Sqcm

6.5.8 K Value (IS 16203) - Equal or less than 0.023W/mk at 10^0 temp & 0.025 w/mk at 25^0 temp

6.5.9 Water absorption (ISO 2986) – 8-5 ng/Pasm(max)

6.5.10 Close Cell Content (IS 11239 Part 5) – Min 85 %

6.5.11 Air and water permeability – No absorption of vapor or water

6.5.12 Horizontal burning characteristics – max 25 mm

6.5.13 Dimensional Stability at 100^0 C and - 40^0 C Overall max 2.0 % (IS 11239:Part 3)

6.5.14 Core Type – Self blended hybrid insulation core

6.5.15 Composition – Foam based closed cell structure

6.5.16 Green Mtrls – 100 % recyclable

6.5.17 Fire certification – Fire safe test report showing integration, insulation as per EN13501-2

6.5.18 Effect of Fire – No loss of insulation core to be ensured

6.5.19 **Smoke generation**

6.5.19.1 Smoke generation rate at 12 min <0.07g/s

6.5.19.2 Max smoke generation rate <0.23 g/s

6.5.19.3 Total smoke generated < 60 g

6.5.20 Joints in Panel – Joint less (Length upto 11.8 Mtr)

6.5.21 **Ash Content**

6.5.21.1 The core material should have minimum ash content of 90 % when tested without adhesive

6.5.21.2 The core material has max heat combustion of 2.0 kj/g (860 BTU/lb) when tested without adhesive

6.5.21.3 The core material should not show visible flaming when tested at an applied heat flux of 50 kW /m2 in air enriched to 40 % oxygen without adhesive or facers.

6.5.22 **Note :- All panels will be manufactured in single piece as per approved layout drawings using the above materials and manufacturing process. Continuous line Polyisocyanurate foam Panels, being used in wall /ceiling /any other location will be tested for its physical /chemical /mechanical properties. Test Certificate for PIR insulation given by CSIR**

-CBRI Roorkee . Certificate will also give out conformity to the following IS – 12436 -1988 (reaff 2022)IS 11239 part 2,4,5 11 & 12 1985- 1988 (reaff 2019) and flammability test certification conforming EN13501-1 or 476 Part 20 to 22 from CSIR-CBRI Roorkee /IAS labs/NABL will be produced by supplier to consignee at the time of supply of each asset.

6.6 2 mm thick MS Sheet 'C' channel of size 55 mm x 40 mm at the bottom (Floor level) and Top shall be provided to side and fit the wall panels. Suitable arrangement will be made to fix the 'C' channel to floor at bottom and top to structural members.

6.7 All panels should be moulded in place using the above in-situ process through continuous line manufacturing process after placing them in a hydraulic process. A spare panel will be supplied along with bulk supply of panels for testing the designed parameters of the incorporated panels in the shelters.

6.8 The purchaser can get quality testing of any panel from the lot supplied to ensure quality control as per given specifications. Cost of quality testing will be borne by supplier.

7. **Locking arrangement**. All PIR panels will be provided with suitable number of cam-lock which are fixed in place during the in situ process itself. The placing of cam-lock should be eccentric and such that a 1.2 m long x 1.2 m wide wall panel should be provided with three pairs of cam-lock of the vertical joints. The holes in the panel for operating the cam-lock should be provided with PVC caps for sealing after installations.

8. **Gable End**. PPGI metal skin PIR panels of 50 mm. shall be provided at both gable ends excluding the stair case of the shelters with necessary fixing arrangement (factory Cut) same as wall.

9. **False ceiling**. The false ceiling will be provided in the shelter (**including verandah**) both in first floor & ground floor. Details are as given below :-

(9.1) The false ceiling shall be made of 40mm thick PIR panels. The Panels will be made of 0.5mm thick PPGI on both side with 50mm thick layer of rigid CFC free polyurethane foam of density 40 kg +/- 2kg/ cum as Insulation. False ceiling panels are to be manufactured in single length & shall be installed covering the entire width of the living area resting on wall panels on either side. The false ceiling panels will have tongue & groove arrangement & shall be interlocked using cam locks systems. MS 'T' of 75 mm x 75 mm x 10 mm shall be provided to lay ceiling panels as per **Drg' DSB- 03** attached. Aluminum powder coated 'L' flashing of size 40 mm x 40 mm x 3 mm shall be used to fix all around the joining point of wall panel & false ceiling and vertically from floor level to the false ceiling level joining the walls at the corner junction for support & joining by means or rivets at minimum distance of 200 mm on both the faces.

10. **<u>Doors</u>**.

(10.1) **Door size**. 950 mm x 2100 mm.

(10.2) **Specifications of doors are as under** :-

(10.2.1) Frame shall be made from aluminum frame using 2 mm thick aluminum powder coated section confirming to IS 513 1986

(10.2.2) It will also have 04 numbers 100 mm steel hinges welded into the frame.

(10.2.3) Shutter shall be made out of PIR panel of 30 mm thick fixed with aluminum frame 2 mm thick single leaf with fixed arrangement. The main doors shall be provided with fly proof shutter on outside of the door in addition to shutter specified earlier. The galvanized stainless steel wire mesh shall be of stainless steel and average aperture 1.18 mm.

(10.2.4) Shutter frame shall be made of aluminum sheet of 105 mm x 60 mm powder coated.

(10.2.5) Fly proof shutter will be openings in the verandah whereas main shutter will be opening inside the shelter.

(10.2.6) Door shutter be fixed with 02 Nos aluminum aldrop of 300 mm length, aluminum tower bolt of size 200 mm at the top and 100 mm at the bottom on each shutter, one sliding bolt 300 mm long and one No 1500 mm long bow handles with fixing arrangement.

11. **Windows**.

(11.1) All windows shall be provisioned at sill height of 900mm. Window shall be of size 1000mm x 1000mm. Window shall be triple track sliding type made of **UPVC white / off white** colour (Make : Anglian, Everest, Duraflex, Quick slide) of size 63mm x 38mm x 2mm frame with 4mm thick glass pane with two shutter glazed and one shutter gauged third shutter shall be having stainless steel fly mesh of 0.56mm thickness and 1.4mm aperture complete in all as directed.

(11.2) Necessary locking arrangement shall be made for the window shutters.

(11.3) The wall panel housing the window to be precut to the window specification ex-factory and as per the spacing shown in the drawing attached.

12. **Ventilators.**

(12.1) Aluminum ventilators of size 1000 mm x 450 mm with two glass panels and fixed steel wire mesh on outer side to be provided. The ventilator frame will be manufactured using suitable Hindalco/ Jindal aluminum powder coated sections. The ventilator shall have double glass of 5mm thick for full length of 1000mm and width of size 300 mm, outside one fixed from top and inner side one fixed from bottom so that there is staggered gap in between them for air ventilation. The fly proof shall be of stainless steel wire mesh with 1.18 mm aperture fixed on outer frame work. The ventilator frame should be so designed that it gets firmly fixed with wall panel.

(12.2) Suitable arrangement shall be made to remove the glasses to enable cleaning of glasses and also the wire mesh

(12.3) The wall panel housing the ventilator to be precut to the ventilator specification ex factory

(12.4) The No of ventilators are same as of windows, fixed exactly above and in line with the windows, 300 mm from the false ceiling level.

13. **Sunshades**. Suitable sunshades at the slope of 20 degree downward will be provided for all external doors and windows not covered in verandah. Material used for making sun shade shall be pre coated pre profile steel sheet of 0.6 mm thick on angle iron frame 25 mm x 25 mm x 3 mm with fixing arrangement. Pre coated steel sheets shall match with the colour of the outer colour of wall panels. The minimum projection of the sunshade shall be 450 mm and 300 mm wider than the width of the openings.

14. **Finishes.** All steel work shall be given two coats of red oxide zinc chromite primer by the manufacturer in the factory. All exposed steel members shall be given two coats of colour paint as approved by the Commanding Officer, 116 Engineer Regiment.

15. **Foundation**. Foundation drawings of the structures and foundation bolts, nuts and all associated accessories shall be provided & constructed as per drg and direction by Engr-in-Charge.

16. **<u>Flooring</u>**.

(16.1) **<u>Ground floor</u>**.

(16.1.1) 150 mm thick sub base of hard core of broken stone aggregate.

(16.1.2) 100 mm thick base course in PCC 1:4:8.

(16.1.3) 100 mm thick layer of PCC 1:2:4, type BO over base course.

(16.1.4) 1200 mm x 600 mm x 10 mm thick non skid ceramic digital Vetrified tiles shall be provided over 20 mm thick screed bed in CM1:4.

(16.2) **<u>First Floor</u>**. Provision of material and flooring of the first floor shall be carried out by the firm. The specification for the flooring shall be as under :-

(16.2.1) MS decking plate of 1.0 mm thick.

(16.2.2) 100 mm thick reinforced concrete 1:11/2:3 over steel decking all as per design. The design to be vetted by NIT and approved by Commanding Officer.

(16.2.3) 600 mm x 600 mm x 10 mm thick non skid ceramic Vetrified tiles shall be provided over 20 mm thick screed bed in CM1:4.

17. **Dwarf Wall for Plinth.** Dwarf will be done as per drg and direction by Engr-in-Charge.

18. **Plinths**. Plinth will be done as per drg and direction by Engr-in-Charge.

19. **Foundation of Column.** Foundation for the column shall be constructed as per the drawings att and four Nos of foundation bolts 685 mm long 20 mm dia with nuts for each column will be provided.

20. **Eve Board**. Bird board of size minimum 30 cm width will be provided of pre coated plain sheet of 0.6 mm in Z section shape. Colour of Bird Board should be matching with the colour of the wall panel.

21. **Curtain Rods with Fixing Arrangements**. Curtain rod arrangements shall be provided on all doors and windows. The specification for the same is as under :-

(21.1) Mtrl	-	Mild steel powder colour coated fancy type drapery rod
(21.2) Size	-	25 mm dia.
(21.3) Over hang projection	-	450 mm on both sides.
(21.4) Fixing arrangement	-	Load of the rod to be on main columns.
(21.5) Brackets for holding the rods	-	Adequate number to support the rod.

22. **Gap Filling**. Minimum 15 Nos Silicon Tube for gap filing shall be provided with 2 x Silicon machine.

23. **Workmanship**.

(23.1) Welded connection will be provided conforming to IS-806-1968.

(23.2) Fabrication provisions in section 11 of IS-800 -1984 will be apply to all types of steel being used for fabrication.

(23.3) Various fasteners and fittings of mild steel shall be supplied unless otherwise Specified in and shall be fixed where required. Fastening means will be provided for the items to be fitted at the time of erection. 10% spare nuts, bolts, screw, hooks and washers etc will be provided per shelter. The fasteners supplied shall be conform to relevant BIS specifications.

24. **Electrification**. Electrical points for lights/fans/power shall be provided as given in the store list attached in **Annx–II**. In addition main board/MCB DBs comprising of SP/DB MCBs shall be provided. All electrical items (MCB/switches/Cables etc) shall be ISI marked and confirming to relevant IS specifications.

25. **Fan Hooks**. Hooks for fixing of ceiling fans are to be provided with supporting rod connected with the 'T' joint supporting the false ceiling panels with proper arrangements for securing it with help of nuts and bolts along with safety locking pin.

26. **Minor Details**. Details not mentioned in the specifications but required as per sound and good engineering practice shall be provided by the supplier. Only superior specifications shall be accepted, after obtaining due clearance by the Commanding Officer.

27. **Material**. All materials required for manufacture of the shelter shall be new and Materials shall comply with relevant bureau specifications and bear the mark.

28. **References of drawings**. Drawing att.

29. Various members and their quantities to be provided with each shelter as given drawings enclosed is tentative and an indicative only. The exact quantities to be provided with each shelter shall be as per actual requirement.

30. **Materials Make** :- The quality and sizes of the various types of items provided or used while supply or fabricating shall be conforming to relevant technical specifications. **The make that are offered will be intimated for approval of Accepting Officer within 07 days from date of placing of supply order.** Once make approved as offered and accepted will be final and all materials supplied under supply order will conform to the approved make offered. In case of any disputes / missing of details on account of specifications / makes, bidders are requested to intimate to tender inviting officer before bid quoting. Once tender is accepted no deviation shall be entertained. In this case decision of Accepting Officer shall be final and binding.

31. **Quality Control**. Accepting officer is free to get all parts/any part checked for quality conformation as given in the technical specifications irrespective of DGQA inspection. Cost of testing will be borne by the supplier.

NOTE: 1. The supplier will be whole and sole responsible to supply the items and materials required for construction as per specifications and good engineering practice including minor extras.

Station : C/o 99 APO

Dated 22 Jun 2026

(Ajay Londhe)
Brig
Chief Engineer

Appx'B'
(Ref Ser No 01 of Costed Sch to TS No 1 of
Job No OT/EC/3/56/2026-27/ 2655

TECHNICAL SPECIFICATION OF OR LIVING SHELTER D/S 18.0 X 6.10 M WITH 1.50 M WIDE VERANDAH

1. **Foundation**. PCC foundation will be made by using 400mm x 200mm x 200mm pre casted concrete blocks, in cement mortar 1:6 over 100 mm thick PCC 1:4:8 type D2 by using 40mm graded stone aggregate over 75mm thick 3" graded stone aggregate stone hard core all as shown in drg No 01 of 01 (PCC fdn details) att. 15mm thick cement plaster in CM 1:4 will be done on block masonry above ground level on the external surface.

2. **Column**. 18 Nos of columns, 10 No Columns of size 550mm x 330mm and 8 Nos of columns of size 500mm x 300mm by using RCC 1:2:4 type B1 using 20mm graded crushed stone aggregate. The size of footing – I & Footing – II will be 1500 x 1600mm and 1400 x 900mm over 75mm thick PCC 1:4:8 type D2 over 75mm thick hardcore over compacted soil. Complete work to be executed as per drg details att.

3. **Plinth Beam**. Plinth beam will be of size 300mm x 230mm using RCC 1:2:4 (20mm graded aggregate). 06 x 12mm TMT Bar be used for reinforcement and 8mm TMT bar will be used as stirrups at a distance of 200 c/c over 75mm thick PCC 1:4:8 40mm graded aggregate.

4. **Flooring:-**

(4.1) **Ground floor**.

(4.1.1) 150 mm thick sub base of hard core of broken stone aggregate.

(4.1.2) 100 mm thick base course in PCC 1:4:8.

(4.1.3) 100 mm thick layer of PCC 1:2:4, type BO over base course.

(4.1.4) 1200 mm x 600 mm x 10 mm thick non ceramic colour Digital tiles shall be provided over 20 mm thick screed bed in CM 1:4.

(4.2) **First Floor**. Provision of material and flooring of the first floor shall be carried out by the firm. The specification for the flooring shall be as under :-

(4.2.1) MS decking plate of 1.0 mm thick.

(4.2.2) 100 mm thick reinforced concrete 1:1 1/2:3 over steel decking all as per design. The design to be vetted by NIT and approved by Commanding Officer.

(4.2.3) 600 mm x 600 mm x 10 mm thick non skid ceramic tiles shall be provided over 20 mm thick screed bed in CM1:4..

5. **Plinth Protection**. All around the shelter, plinth protection of 750 mm wide, 75mm thick with PCC 1:3:6 type C1 shall be laid over 75mm thick hard core over compacted soil.

6. **Steps**. A step shall be constructed out of PCC sold brick masonry in CM 1:6 in front of the verandah in alignment with riser as 200mm and treads as 300mm.

7. Store list of constr mtrl att as **Annx 'I' & Annx-II**

8. **Materials Make** :- The quality and sizes of the various types of items provided or used while supply or fabricating shall be conforming to relevant technical specifications. **The make that are offered will be intimated for approval of Accepting Officer within 07 days from date of placing of supply order**. Once make approved as offered and accepted will be final and all materials supplied under supply order will conform to the approved make offered. In case of any disputes / missing of details on account of specifications / makes, bidders are requested to intimate to tender inviting officer before bid quoting. Once tender is accepted no deviation shall be entertained. In this case decision of Accepting Officer shall be final and binding.

9. **Quality Control**. Accepting officer is free to get all parts/any part checked for quality conformation as given in the technical specifications irrespective of DGQA inspection. Cost of testing will be borne by the supplier.

NOTE: 1. The supplier will be whole and sole responsible to supply the items and materials required for construction as per specifications and good engineering practice including minor extras.

(Ajay Londhe)
Brig
Chief Engineer

Station : C/o 99 APO

Dated : 18 Jun 2026

Appx 'C'
(Ref to Ser No 2 of CS to TS No 01 of
Job No OT/EC/3/2026-27/2655

DETAILED TECH SPECIFICATION FOR PATHWAY

1. **Earth compaction.**

(1.1) Forming embankments including raising or lowering of earth, spreading in layers, watering ramming/rolling.

(1.2) The compaction shall be done with the help of roller of 80 to 100 Kilo Newtons static weight with plain or pad foot drum or heavy pneumatic tyred roller of adequate capacity capable of achieving required compaction.

2. **Granular Sub Base (Soling).** Sub base shall be executed using GSB 150mm thick with stone aggregates/broken stone mixed with sand confirming to grading No 1 as per Table 400-2 of MORT&H and rolled with 8 to 10 Ton roller to required camber and gradient for **150 mm** spread thickness. The crushed or broken stone shall be hard durable and free from excess flat elongated soft and disintegrated particles dirt and other deleterious material.

3. **PCC 1:4:8.** **75mm** thick Plain Cement Concrete (PCC 1:4:8), Type D2, with using 40 mm graded stone aggregate as in sub base course. The crushed stone aggregate shall be hard durable and free from excess flat elongated soft and disintegrated particles dirt and other deleterious material.

3. **Sand Layer.** **40MM** Thick Sand Layer Cushion to be laid all over the track before laying of interlocking bricks. Sand should be free from excess flat elongated soft and disintegrated particles dirt and other deleterious material and should confirm to the grading of coarse sand. Test certificates in respect of bulking and moisture content should be submitted to this office before supply of material from approved Govt Lab and should be within permissible limits.

4. **Interlocking PCC Paver Blocks**

(4.1) Interlocking Paver Blocks **230 x 115 x 60 mm** minimum strength M35 Grade Concrete. Precast Block shall conform to IS-15658 2006. Solid Block made out of PCC 1:3:6 having the specified compressive strength of paver blocks at 28 Days should be 35 KN/Sqmm.

(4.2) The blocks are to be aid in herringbone pattern as attached in drawing.

(4.3) After laying of blocks the gaps is to be filled with fine sand and compaction of blocks is to be done by using plate vibrator.

(4.4) The shoulder of pathway to be laid with kerb stone size 300x300x100 mm with paint yellow and black .

6. **Additional Points.**

(6.1) Compressive strength of paver block as per IS code is to be submitted before laying of blocks.

(6.2) Complete all as specified in Drg's.

NOTE: 1. The supplier will be whole and sole responsible to supply the items and materials required for construction as per specifications and good engineering practice including minor extras.

Station: c/o 99 APO

Dated : 18 Jun 2026

(Ajay Londhe)
Brig
Chief Engineer

Appx 'D'
(Ref Ser No 3 of CS to TS No 01 of
Job No OT/EC/3/56/2026-27/2655

TECHNICAL SPECIFICATIONS FOR SINGLE SLEEPING BUNK WITH COIR MATTRESS

1. **MS Rectangular Hollow Section**. Mild steel rectangular hollow section shall comply with IS-4923:1997 specification. Rectangular hollow section fitting shall be clearly finished straight and free from scale, crack, surface, lamination and other defects. MS Rectangular hollow section including fittings of size as shown in drawing used for manufacture shall be as per approved design and drawing to provide adequate strength and life of single sleeping bunk.

2. **Plywood Base.** Plywood base of size **1981.20 mm x 920mm x 12mm** thick of synthetic resin Bonded Grade Boiling Water Resistant (BWR) type, termite resistant and treated with preservative as per IS–5539-1969 shall be used as base over the iron frame for laying of mattress. The plywood will be fixed to the frame with flat head self tightening screws.

3. **Nuts and Bolts**. Hexagonal bolts and nuts will be provided as per design requirement conforming to IS : 1363-1967 shall be used for assembling of individual members. Allen key bolts/flat headed self tapping bolts shall also be used as required 10% extra quantity of all nuts, bolts and washers will be supplied to cater for losses and breakages in transits.

4. **Welding**. Proper welding joint will be formed where specified as per the design. All welding works will be as per IS: 823 – 1964 and IS : 816 – 1976. Welding shall be done by MIG welding process and shall be free from cracks and defects.

5. **Finishes**. The bed bunks shall be painted with two coats of anti-corrosive red oxide primer at the time of supply. Before application of paint all the steel surface shall be thoroughly cleaned to make the same free of rust, mill scale, dirt, oil etc either by mechanical or chemical means. The bed bunks should be painted with two coats of silver grey paint at factory or on site after assembling. Similarly steel locker will be painted with two coats of enamel paint, colour as specified by the consignee over a coat of anti-corrosive primer applied through spray painting.

6. **Coir Mattress**. Coir mattress of size **1981mm x 900mm x 75mm** thick comprising of 50mm thick high density coir with hessian border pasted with 25mm thick high density foam (pasting to be done with dendrite adhesive) Mattress to be stitched with good quality OG cotton fabric or as colour approved by Commanding Officer with double stitching on the margin/edge all as per sample approved. **Make: Sleep well/Kurl-on/Duroflex**

7. **Minor Details**. Details not mentioned in the specifications but required as per sound and good engineering practice should be provided by the supplier.

8. **Reference of Drawings**. Drawing No 1/1 of Single Sleeping Bunk.

Station : C/o 99 APO

Dated : 16 Jun 2026

(Ajay Londhe)
Brig
Chief Engineer

Appx 'E'
(Ref Ser No 3 of Costed Sch to TS No 01 of
Job No OT/EC/3/56/2026-27/ 2653

TECHNICAL SPECIFICATIONS FOR BED SIDE LOCKER (SMALL)

1. **Bed Side Locker**. Bed side locker of size 900mm x 500mm x 1000mm made out of 20 gauge CR steel sheet conforming to IS : 513:1994 amdt 1 & 2. The locker will have two compartments with two separate doors on the front side and a partition on the inner side in the middle dividing the width of the locker into two equal parts. Left Compartment will have two shelves placed horizontally at equal distance and right compartment will have one 20mm dia steel rod 65mm below the top level for hangar arrangement. Vertical reinforcement of 800/400mm to be fitted inside on both the doors as well as on right side of the locker (Hanger Side).

2. **Painting**. Two coats of enamel paint dark OG colour/as indicated by consignee over a coat of anticorrosive primer applied through spray painting. Before application of paint all the steel surface shall be thoroughly cleaned to make the same free of rust, mill scale, dirt, oil etc either by mechanical or chemical means.

3. **Lock**. The locker shall have Hasp & staples as locking arrangement and two steel handles fitted on the front doors. Two tower bolts of 100mm to be fixed inside the left door on top & bottom for locking arrangement. For restricting the move doors inside, two stoppers shall be provided one on the top for full width of the locker and second at the middle of the bottom (1400mm) all as shown in the drg.

4. **Base**. The locker will be placed on 2 Nos, 100mm high hollow base made with M.S sheet channels.

5. **Minor Detail** Minor details not mentioned separately in the specifications but required as per sound and good engineering practice shall be adhered to.

6. **Reference of Drawing**. Drawing No 1/1 of Bed Side Locker.

Station : C/o 99 APO

Dated : 18 Jun 2026

(Ajay Londhe)
Brig
Chief Engineer

Appx 'F'
(Ref Ser No 3 of Costed Sch to TS No 01 of
Job No OT/EC/3/56/2026-27/2653

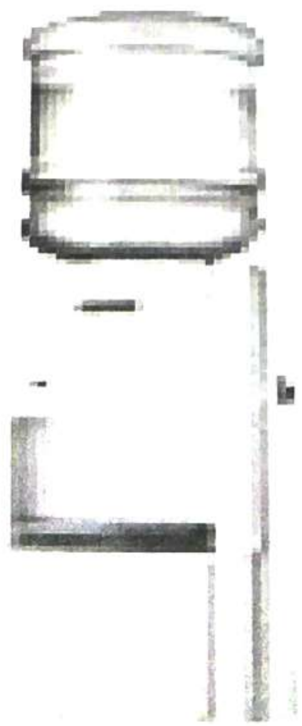
TECHNICAL SPECIFICATIONS FOR FURNITURE

1. **Chair Writing**. Chair writing of size **535mm x 470mm x 890mm** shall be provided. Chair shall be made out of well seasoned 1st class teak wood free from knots. All wooden exposed surfaces shall be finished with two coat of French polish high gloss finish after preparation of wooden surfaces. Plastic cane seat of size 470mm x 470mm x 535mm with arm rest of size 60mm x 25mm shall be provided. The legs will be fitted with hard PVC shoe. Refer Drg No FD 280 sheet 1/1.

2. **Writing Table**. Table of size **1145mm x 610mm x 760mm** shall be provided. Table top shall be 19 mm thick pre-laminated decorated particle board fixed with hollow section frame with screw @ 200 mm c/c. The frame work of table shall be of MS hollow section of size 25mm x 25mm all as per drg. End of legs will be sealed with MS sheet 1.6 mm thick with proper welding process to avoid the dust entry. 6 mm thick teak wood lapping shall be provided on all the exposed edges of MDF duly treated with French polish as specified. All steel surfaces will have two coats of synthetic enamel paint over a coat of red oxide primer. The legs will be fitted with hard PVC shoe. Refer Drg No WT 1/1.

3. **Bottled Water Dispenser**. Bottled Water dispenser body made of ABS plastic and tank made of Food Grade stainless steel. Capacity of cooling and hot water shall be of 3 liters and 5 liters respectively with 3 temperature taps like Hot, Plain and cold. Heavy duty bottle piercer with bottle holder shall be provided to hold the water bottle of capacity 20 ltrs. Bottom of the body shall consist of a refrigerator of 14 ltrs capacity. Make: Blue star / Usha.

WATER DISPENSER



4.0 **Powder based Fire Extinguisher (Red) - 1** : Dry powder fire extinguisher of 1 Kg Capacity in confirmation with ISO/CE certified with wall mounting holder. It shall be helium leak tested and with 3 years warranty. Make : Ceasefire / Suprema / Eco fire

(Ajay Londhe)
Brig
Chief Engineer

Station : C/o 99 APO

Dated : 16 Jun 2026

Annx 'I'
(Ref Ser No 01 of CS to TS No 01 of Job No OT/EC/3/26-27/ 2655

STORE LIST OF CONSTR MTRL FOR OR LIVING SHELTER (D/S)

Ser No	Description of Stores	A/U	Qty	Remarks
1	Shelter Part only	Set	1	
	Construction Material			
1	Cement :Portland pozzolana cement : PPC Cement IS-1489 (part-2) for clained clay based PPC and IS 1489 (part-1) for fly ash based PPC) packed in 50 Kgs HDPE bag conforming to IS 11652/2000 **Make : Ambuja/ ACC/ Ultra tech/ Dalmia/ Ramco/ Birla/ CCI)**	Bags	615	
2	**Coarse Sand** : Sand shall be clean, sharp, angular hard and durable free from adherent coating and shall not contain clay, mica and other impurities. The maximum quantities of clay and fine silt in sand shall not be more than 5% by mass as per IS-383-1970.	Cum	45	
3	**20mm Stone Aggregates** : Stone aggregate shall be crushed stone 20mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	38	
4	**40mm Stone Aggregates** : Stone aggregate shall be crushed stone 40mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	22	
5	**Hardcore 63 mm Stone Aggregates** : Stone aggregate shall be crushed stone not exceeding 63mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	24	
6	PCC Solid block 400x200x200mm (+/-10mm) minimum strength 50Kg/Sqcm. PCC shall conform to IS-2185 (Part I) 2005. Solid Block made out of PCC 1:3:6 having the compressive strength 5 KN/Sqmm.	Nos	632	
7	Non skid ceramic Coloured Digital tile of size **1200mm x 600mm** x 9-10 mm thick (Make:Kajaria/Asian/Orient/Johnson)	Nos	407	
8	White Cement (Make- Birla Gold)	Kg	50	
9	TMT Bar 8mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	330	
10	TMT Bar 10mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	1740	
11	TMT Bar 12mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	350	
12	TMT Bar 16mm dia (Make- TATA/SAIL/JAI BALAJI)	Kg	1030	
13	ACP(Aliuinium Composite Panal) Size 1.50*1.00	Nos	1	
14	Stainless steel 300 mm long Word	Nos	10	
15	Water proofing compound (Make- Dr. Fixit)	Kgs	8	
16	Lime for white wash	Kgs	15	
17	Binding Wire 0.90mm	Kg	20	
18	Ply wood 12mm thick size 8' x 4'	Nos	12	
19	Balli 100mm dia 3 mtr	Nos	40	
20	Wooden runners 2" x 2"	RM	250	
21	Nails	Kg	20	

	Items			
	LC set with safe chemical 25.1 Zinc & Alloy Coated Safe Earthing Electrode (76.3mm dia) 2 mtr long with back frilled compound conductotile - 2 Bags CPRI approved 25.2 Air Terminal single pointed aluminium rod 12mm dia and 600mm. 25.3 GI pipe med cgraded 12 mtr loing 40mm dia. Make TATA/JINDAL/BANSAL 25.4 Insulator with screw 20 Nos 25.5 Aluminium cable 70Sqmm multi standed single core-30 RM . Make PLAZA/HAVELLS/ RALLISON ISI Marked 10 RM. 25.6 Testing point terminal box made out of gun metal of phosphorus of size 75mmx75mmx5mm with screw. 25.7 Nut and bolts 2" x 1/2" - 08 Nos 25.8 Foundation Nuts & bolts 16mm x 400mm - 04 Nos. 25.9 Aluminium lugs 70Sqmm for connection of cable with nut & bolts of suitable size - 02 Nos. 25.10 Earthing pit cover- 01 Nos.	Nos	1	
23	Electric Earthing Set complete with items as given below :- 23.1 GI Earthing plate 600mm x 600mm x 6mm thick with two Nos holes - 01 No. 23.2 GI pipe 20mm light grade for watering - 1.2 Mtr. Make : Tata/Jindal/Prakash 23.3 GI wire 4mm dia - 02 Kgs. 23.4 GI funnel with wire mesh suitable for 20mm dia GI pipe - 01 No 23.5 GI strip (32mm x 6mm x 4.00mtr) 23.6 Common salt - 50 Kgs. 23.7 FRP earthing pit with cover - 01 No. 23.8 GI nut bolt with washer of size 10mm dia 25mm long - 2 Nos. 23.9 Charcoal - 50 Kgs.	Nos	1	

(Ajay Londhe)
Brig
Chief Engineer

Station : C/O 99 APO

Dated : 16 Jun 2026

Annx-II
(Ref Ser No 1 of CS to TS No- 1 of Job No OT / EC/3/56/2026-27/ 2655

STORE LIST OF ELECT ITEMS FOR 01 X LIVING SHELTER (ORL) (D/S)

Ser No	Description of items	A/U	Qty	Remarks
1	Surface and wall mounted LED tube rod 25/20 watt, 1600 lumens output with suitable fitting frame. Frame shall be powder coated with aesthetically designed dual colour ends caps with bipin holder, connector block complete. (Make: - Bajaj / Havells / Philips/Syska/Wipro/ Crompton.	Nos	30	
2	Pixel bulk head fitting in deep drawn anodized aluminium body, epoxy white powder coated and heat resistent prismatic glass cover with specially designed gasket with LED lamp 1X 15 watt. (Make -Bajaj / Havells /Phillips/Crompton.	Nos	4	
3	Surface and wall mounted LED batten with Niano diffuser technology to ensure minimum loss in light transmitted 8 watt, LED fitting including 9 watt tube rod complete with suitable fitting frame complete. (Make :-Cat No BN 550W LED 6-6500 (KLED 1 Ft) of Philips or equivalent in Crompton / Bajaj /Havells	Nos	4	
4	Ceiling Fan 1200 mm sweep with 5 step fan regulator one modular, conforms to IS:2312, ISI marked with double ball bearing with modular fan regulator. Fan hooks with MS clamps, bolts and nuts. (Make:- Fan - Khaitan/Usha/ Bajaj/Crompton Greaves. Modular Fan Regulator : Havells / Legrand/ Polycab / Anchor)	Nos	16	
5	Exhaust fan 300mm sweep, with heavy speed motor and louvers. (Make: – Khaitan / Usha/Crompton Greaves /Bajaj.)	Nos	6	
6	PVC insulated heavy duty 1100 volt grade cable with aluminium conductor 3.5 core and 16 sqmm served with sheathing of PVC tape and overall PVC sheathed, flat or circular. (Make:- Plaza / Paragon/ Polycab/ Usha/ Bajaj / Finolex / Havells	RM	100	
7	PVC sheathed unarmoured stranded copper conductor cable of size 6.0 sqmm dia conform to IS 694/1990 (green in colour of 90 Mtr Roll). (Make: - Plaza / Paragon /Finolex / Havells/Richa)	Roll	1	
8	PVC sheathed unarmoured stranded copper conductor cable of size 4.0 sqmm dia conform to IS 694/1990 (red, black and Green of 90 Mtr Roll)(Make: - Plaza / Paragon /Finolex / Havells/Richa)	Roll	1	
9	PVC sheathed unarmoured stranded copper conductor cable of size 1.5 sqmm dia conform to IS 694/1990 (red/ black in colour of 90 Mtr Roll).(Make: - Plaza / Paragon /Finolex / Havells/Richa)	Roll	10	
10	PVC casing capping 32mm wide, not less than 2 mm thick. (Make: - Polycab/Suncab/Indoplast/ Surya/ Anchor/ Prestoplast .)	RM	100	
11	PVC Junction linker, 75mm x 75 mm (Make : Prestotech/ Polycab/ Suncab/ Indoplast/ Payal/ AKG)	Nos	50	
12	Modular switch piano flush type 6 Amps 240 volts single pole one way. (Make :- Havells/ Henly /Anchor/ Legrand.)	Nos	62	
13	Modular socket outlet multipurpose 5 pin 6 amps flush button type. (Make : Havells /Anchor/ Legrand .)	Nos	30	

No	Description of items	A/U	Qty	Remarks
14	Modular socket Combination multipurpose 6 Pin. 6 to 16 Amps.(Make :- Havells /Anchor/ Legrand)	Nos	4	
15	PVC switch board 8" x10" Conform to IS 14772. Make - Prestoteak/Polycab/Suncab/Indoplast.	Nos	15	
16	PVC switch board 7" x 4" Conform to IS 14772. Make - Prestoteak/Polycab/Suncab/Indoplast.	Nos	8	
17	PVC switch board 4" x 4" Conform to IS 14772. Make - Prestoteak/Polycab/Suncab/Indoplast.	Nos	30	
18	Fan regulator (Make:- Havells/Anchor.)	Nos	16	
19	Ceiling rose, surface bakelite, 65 x 50 mm, 2 to 3 terminals. (Make: - Anchor/Crabtree/ Legrand/ Havells.)	Nos	52	
20	House service bracket (Hook type) complete with 3 m long GI pipe 40 mm dia medium grade, 2 Nos 90 degree bends, MS hooks/clamps and stay or guy.	Nos	1	
21	MCB DB 4 Way SPN, 240 volt conform to IS 13032, (Make: Anchor/ Legrand/ Havells/IndoAsian/Polycab)	Nos	2	
22	MCB DB 8 way SPN. (Make:Anchor/Legrand/ Havells/ IndoAsian/ Polycab)	Nos	2	
23	MCB TPN 63 Amps. (Make: Anchor/Legrand/ Havells/ IndoAsian/ Polycab.)	Nos	2	
24	MCB SPN 40 Amps. (Make: Anchor/Legrand/Havells/Polycab/ IndoAsian)	Nos	4	
25	Insulation tape 1.80 cm width and 7.5 m long (Make: Anchor)	Nos	25	
26	No 8 x 19 CSK screw full threaded	Pkt	5	
27	No 8 x 32 CSK screw full threaded	Pkt	5	
28	No 8 x 50 CSK screw full threaded	Pkt	5	
29	No 8 x 75 Bolts and Nuts full threaded	Nos	10	

(Ajay Londhe)
Brig
Chief Engineer

Station : C/o 99 APO

Dated : 18 Jun 2026

Annx - 'III'
(Ref Ser No 2 of CS to TS No 01 of Job No-OT/EC/3/2026-27/ 2655

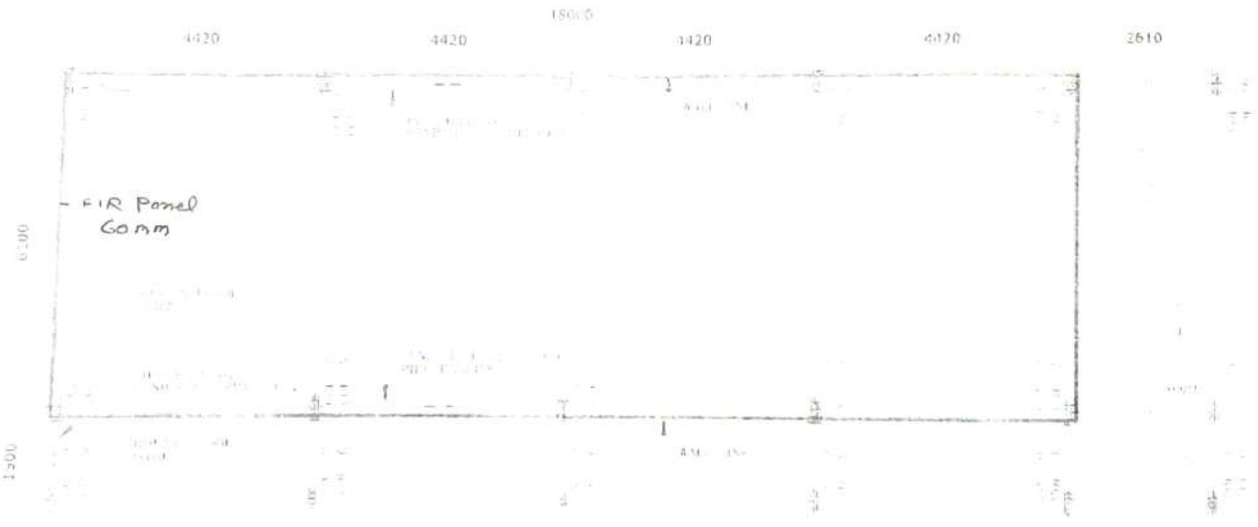
STORE LIST OF CONSTR MTRL FOR PATHWAY (AREA - 12.00 SQM) FOR EACH SHELTER

Ser No	Description of items	A/U	Qty	Remarks
Construction material :-				
1	Cement :Portland pozzolana cement : PPC Cement IS-1489 (part-2) for clained clay based PPC and IS 1489 (part-1) for fly ash based PPC) packed in 50 Kgs HDPE bag conforming to IS 11652/2000 **Make : Ambuja/ ACC/ Ultra tech/ Dalmia/ Ramco/ Birla/ CCI)**	Bags	5	
2	**Coarse Sand** : Sand shall be clean, sharp, angular hard and durable free from adherent coating and shall not contain clay, mica and other impurities. The maximum quantities of clay and fine silt in sand shall not be more than 5% by mass as per IS-383-1970.	Cum	2	
3	**40mm Stone Aggregates** : Stone aggregate shall be crushed stone 40mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	1	
4	**Hardcore 63 mm Stone Aggregates** : Stone aggregate shall be crushed stone not exceeding 63mm graded conforming to IS-383 -1970 and shall be clean hard tough, durable and of uniform quality throughout. The aggregate shall be free from soft disintegrated materials, vegetable and other organic impurities.	Cum	3	
5	Machine pressed precast concrete interlocking paver block of size 225 x 112.50mm confirming to IS 15658-2006 of 60 mm thickness M-35 Grade. And filling the joints with grey cement and coloured pigments as per required.	Nos	479	
6	Providing and laying precast PCC Kerb stone type B1 1:2:4 (20mm graded stone aggregate) of size 300mm x 100mm x 300mm jointing with cement mortar in Cm 1:3.	Nos	70	
7	Kerb stone Synthetic Enamel Paint (Yellow and black Colour)	Ltr	6	
8	Red oxide Primer	Ltr	2	
9	Turpentine Oil (Thinner)	Ltr	2	
10	Painting brush 3" of best quality.	Nos	1	

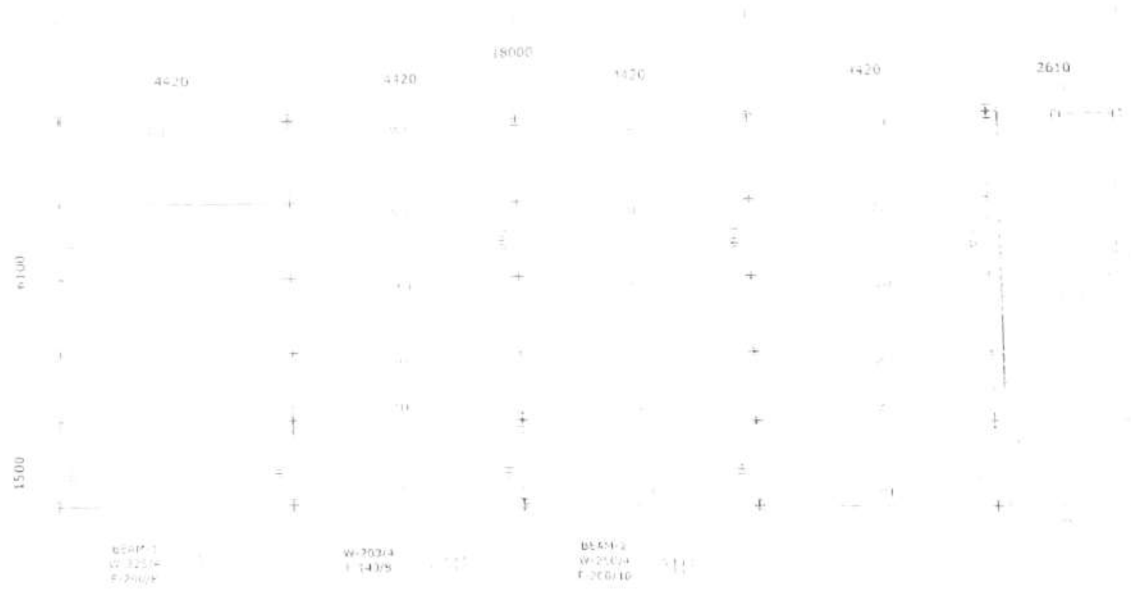
Station : c/o 99 APO

Dated : 13 Jun 2026

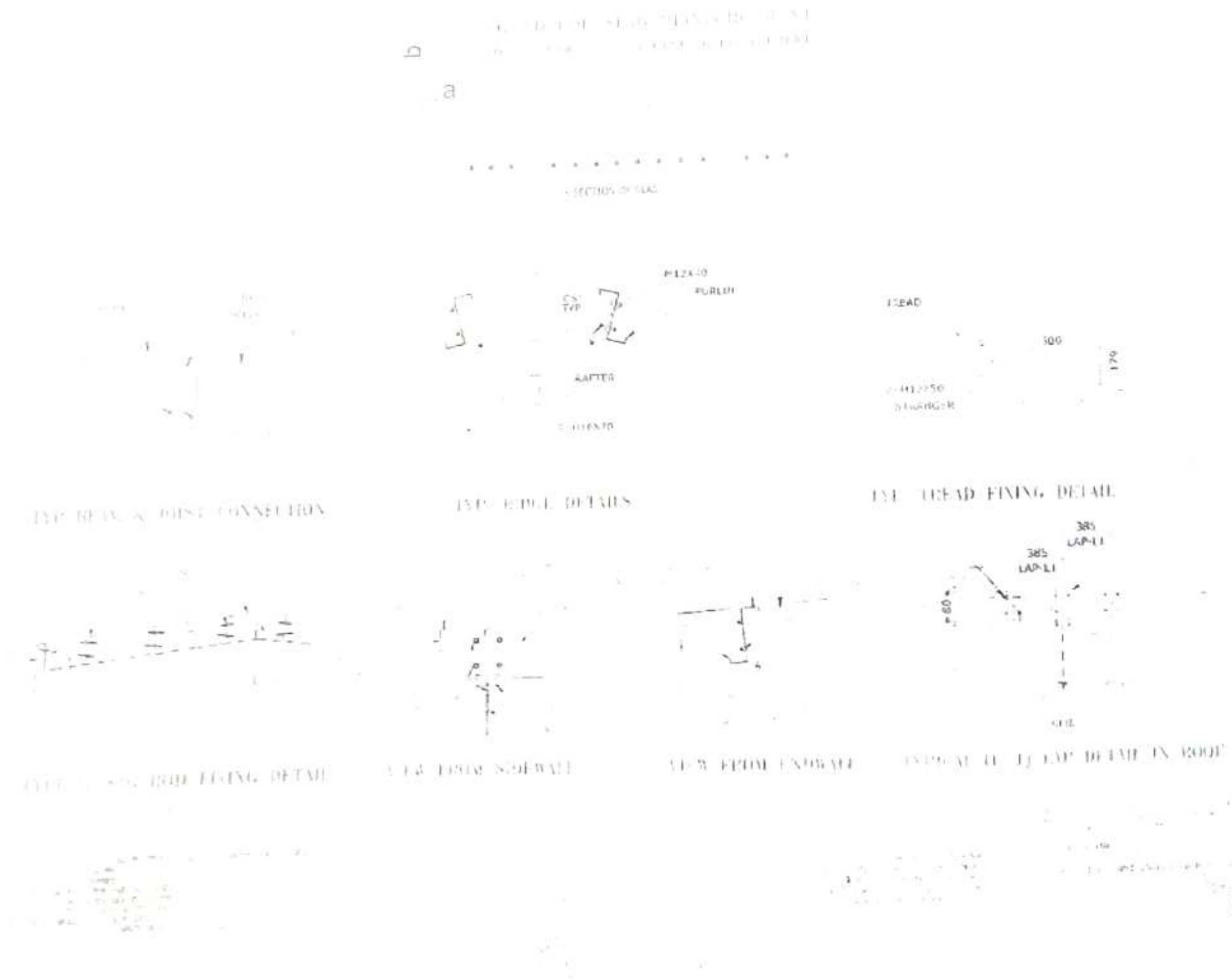
(Ajay Londhe)
Brig
Chief Engineer



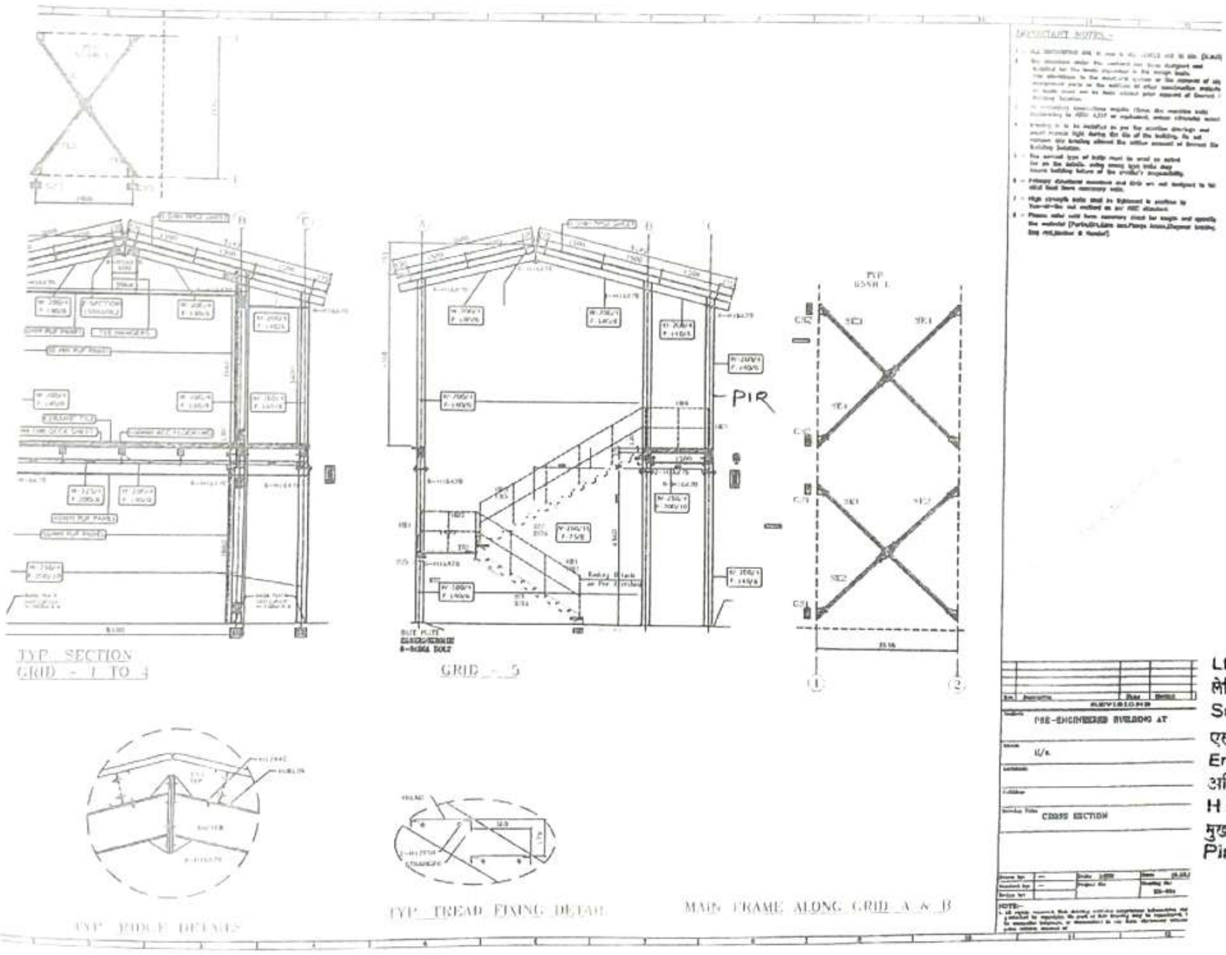
Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता
Engrs Branch
अभियन्ता शाखा
HQ 3 Corps
मुख्यालय 3 कोर
Pin-908503



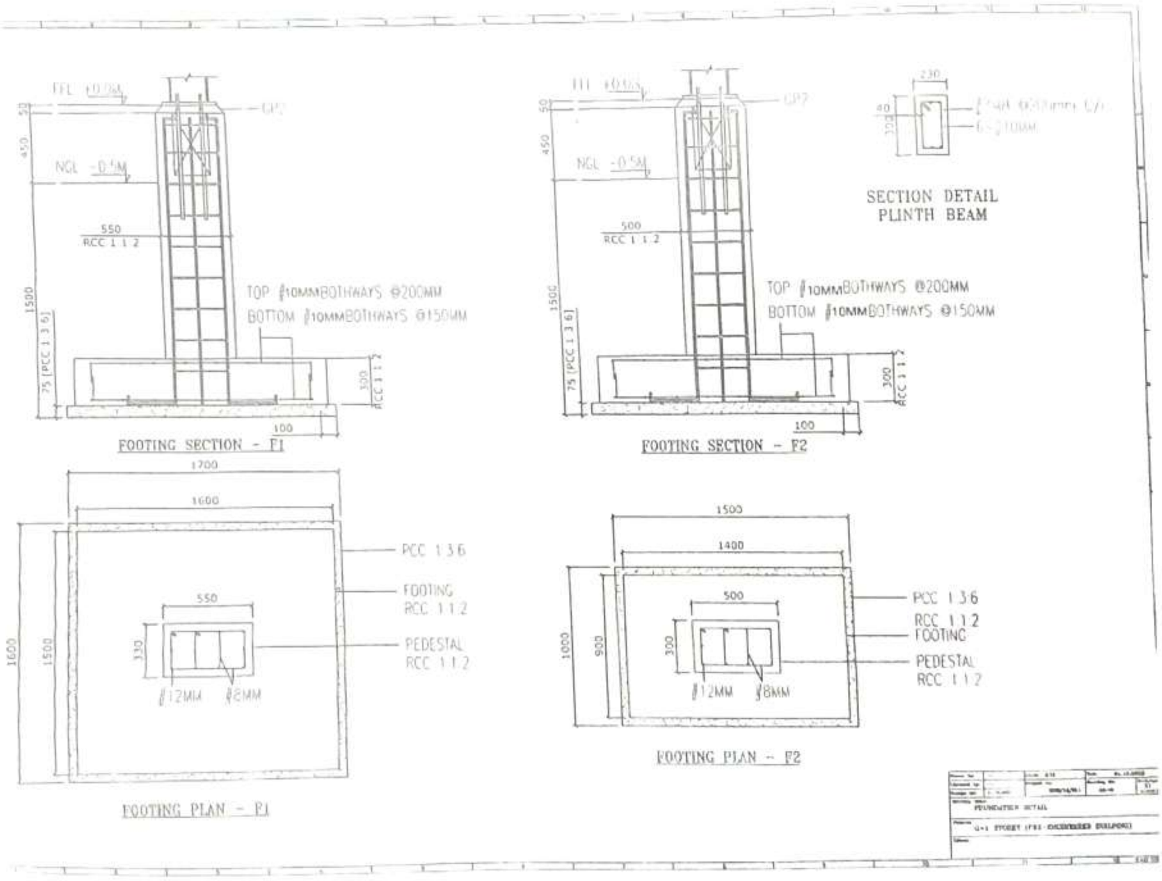
Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
H Q 3 Corps
मुख्यालय 3 कोर
Pin-908503



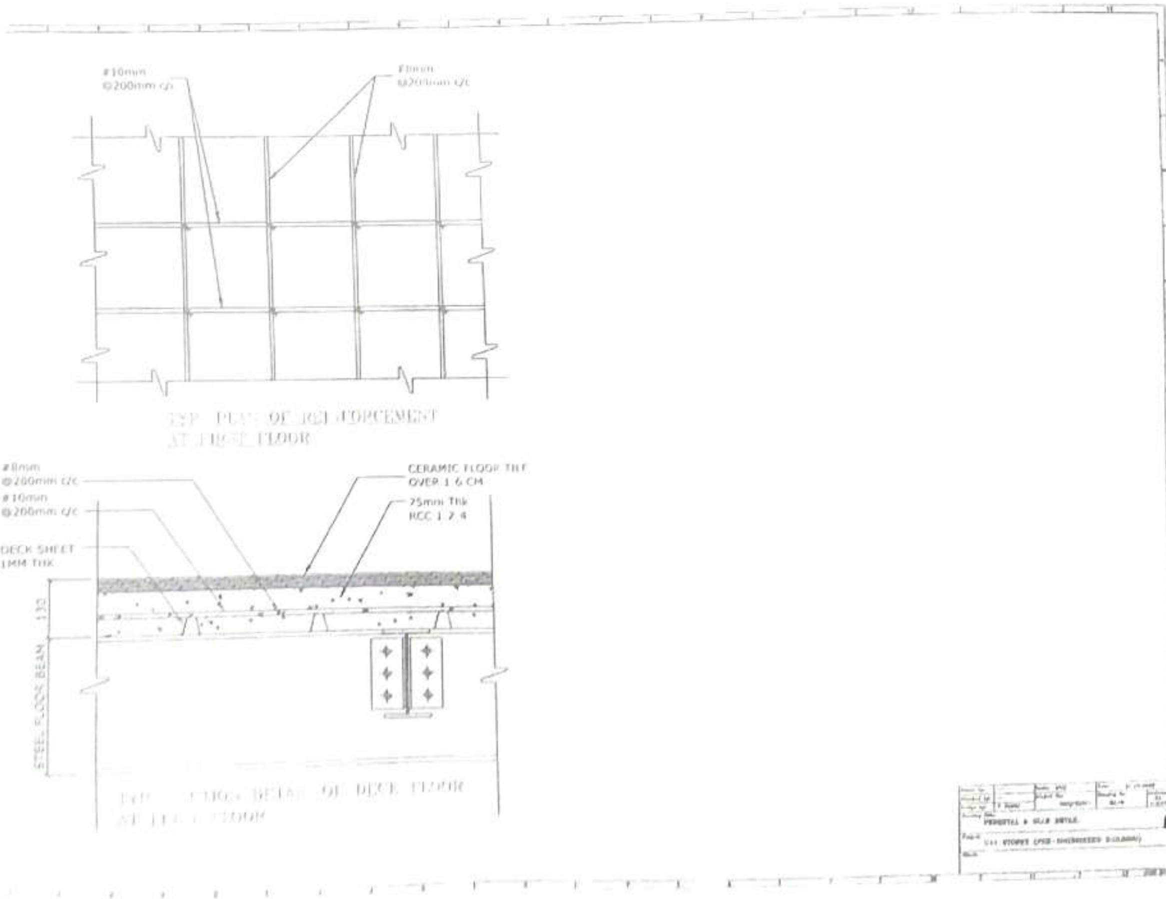
Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
H Q 3 Corps
मुख्यालय 3 कोर
Pin-909593



Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
H Q 3 Corps
मुख्यालय 3 कोर
Pin-908..3

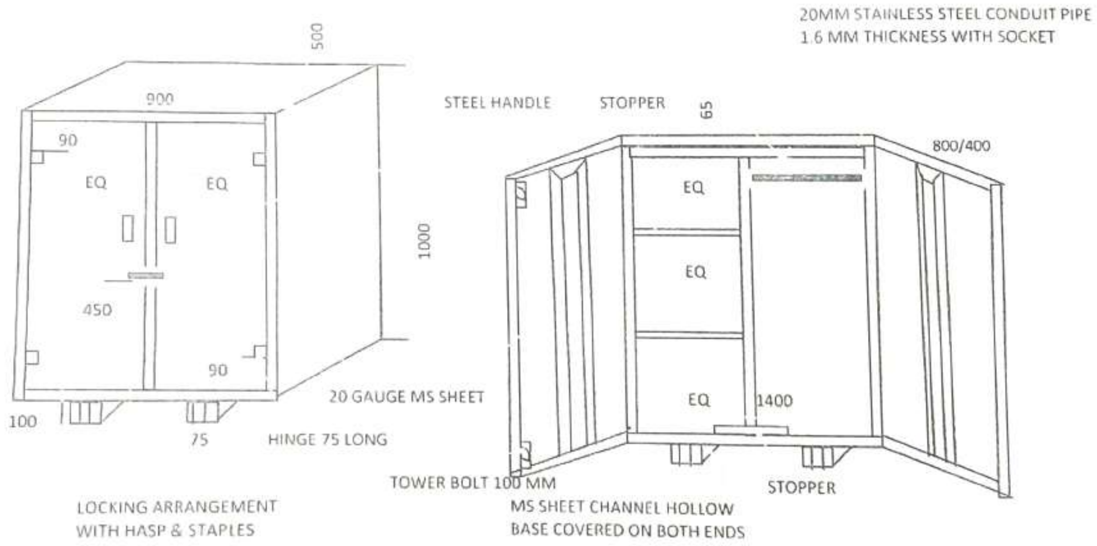


Lt Col
Sol (Engrs)



Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
H Q 3 Corps
मुख्यालय 3 कोर
Pin-90 3

BED SIDE LOCKER

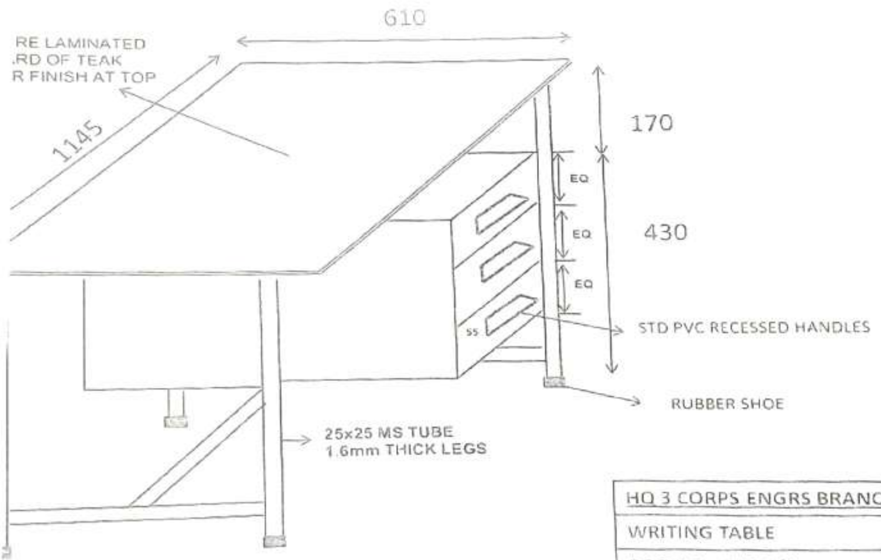


NOTE : ALL DIMENSIONS ARE IN MM
AND DRAWING NOT TO SCALE

H Q 3 CORPS, ENGRS BRANCH
BED SIDE LOCKER
DRG NO : BSL 1/1
GSO 1 (OP WKS)

Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० 1 (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
H Q 3 Corps
मुख्यालय 3 कोर
Pin [illegible] 3

WRITING TABLE



DIMENSIONS ARE IN MM AND NOT TO SCALE

HQ 3 CORPS ENGRS BRANCH
WRITING TABLE
DRG NO : WT 1/1
GSO1 (OP WKS)

Lt Col
लेफ्टिनेन्ट कर्नल
So1 (Engrs)
एस० ओ० १ (अभियन्ता)
Engrs Branch
अभियन्ता शाखा
H Q 3 Corps
मुख्यालय ३ कोर

SINGLE SLEEPING BUNK